End of Study Internship Defense

Optical Characterization for Optimizing Microwave Kinetic Inductance Detector (MKID) for Use in Astronomical Observations

Yann SADOU

09/26/24

Tutors

- Sunil GOLWALA (Caltech)
- Stéphanie LIZY-DESTREZ (ISAE)
- Angélique RISSONS (ISAE)

Caltech

- Private research university in Pasadena, California.
- Focused on Sciences and Technology
  - Space research
  - Sustainability science
  - Quantum physics
  - Earthquake monitoring
  - Soft robotics

NASA Jet Propulsion Laboratory

Team

Physics Mathematics and Astronomy - Observational Cosmology

Sunil Golwala
Principal Investigator
Caltech

Jack Sayers
Assistant Professor - Caltech

Fabien Defrance
Research Scientist - NASA JPL

Jean-Marc Martin
PostDoctoral Fellow - Caltech

Shilling Yu
Graduate Student - UCAS

Sam Condom
Graduate Student - Stanford

Yann Sadou
PostBaccalaureate - Caltech

2 Main Projects

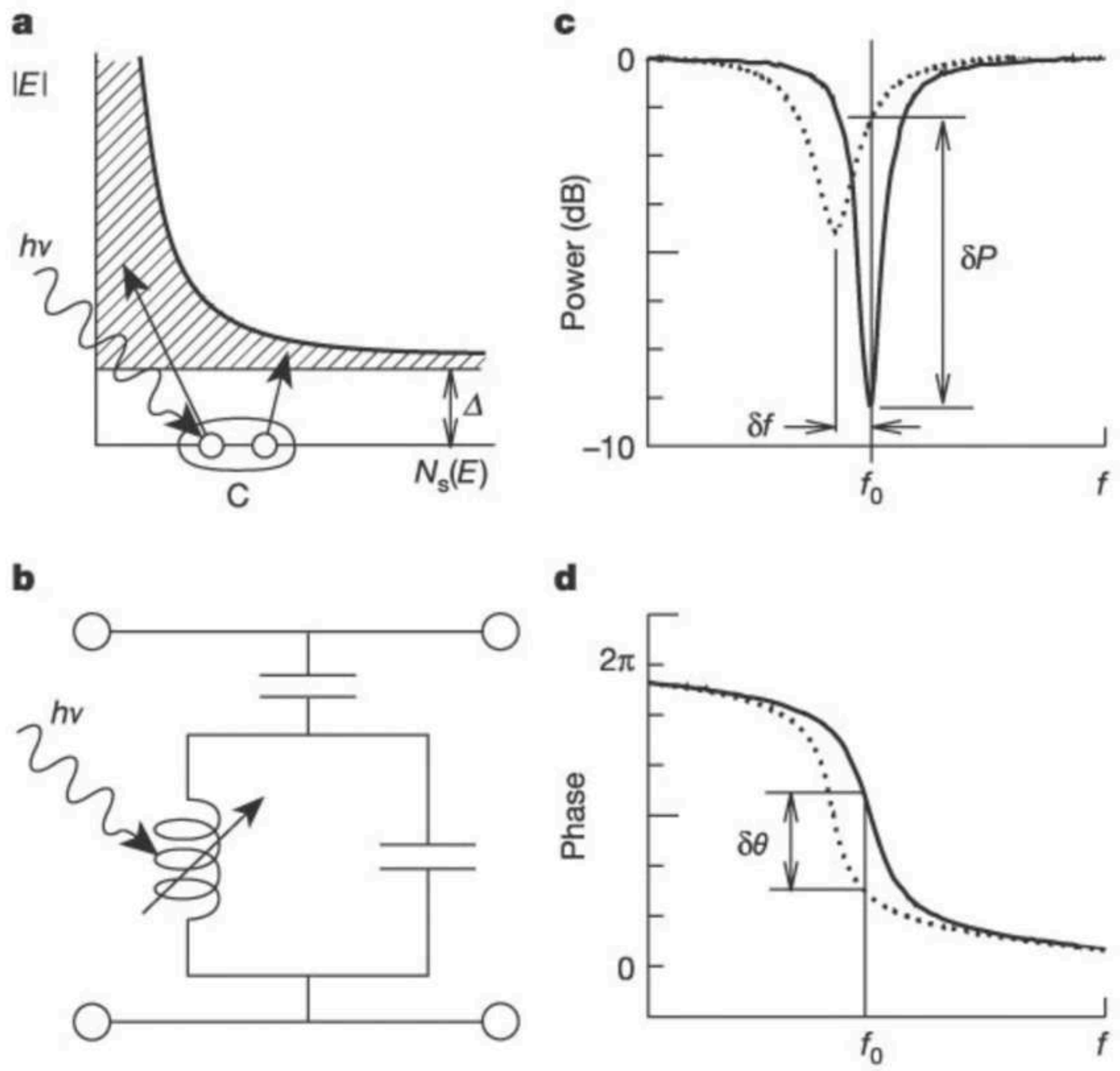
Optical characterization of Microwave Kinetic Inductance Detectors :

Optical simulations, mechanical design of optical structures, construction, measures and data treatment

Developing superconducting phonon detectors for use in direct Dark matter detection experiments:

Cryogenic and mechanical engineering design and assembly of magnetic and blackbody radiation shielding for a dilution refrigerator cryostat in the instrument used to detect dark matter in the universe

Microwave Kinetic Inductance Detector (MKID)



Inductance of the resonator: $L = L_K + L_G$

Resonance frequency: $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$

Breaking Cooper pairs: $N_{qp} \approx \eta \dfrac{h\nu}{\Delta}$ with $E = h\nu$

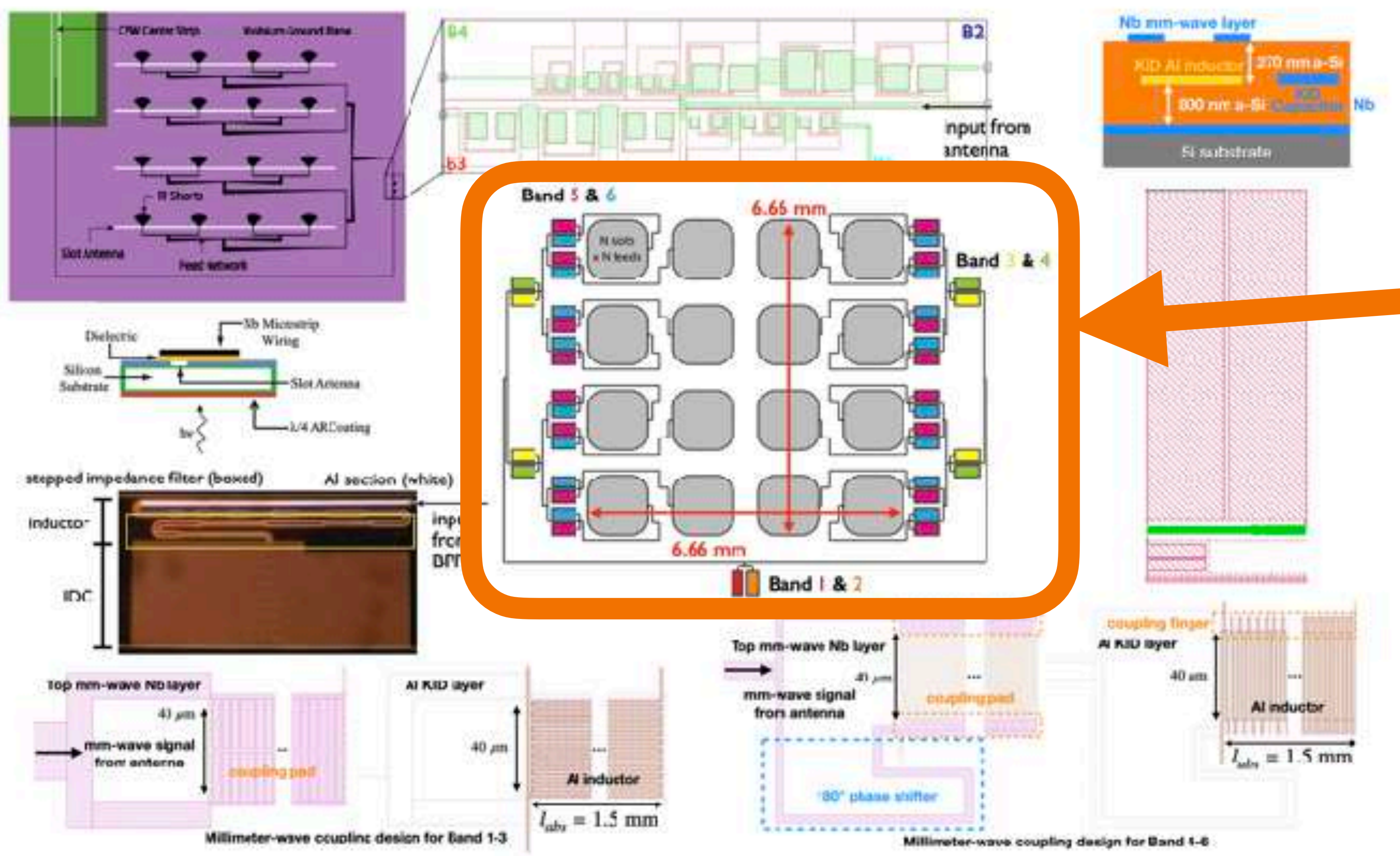
Resonance frequency: $\boxed{\dfrac{\delta f}{f_0} = -\dfrac{\alpha \delta L_K}{2L_K}}$

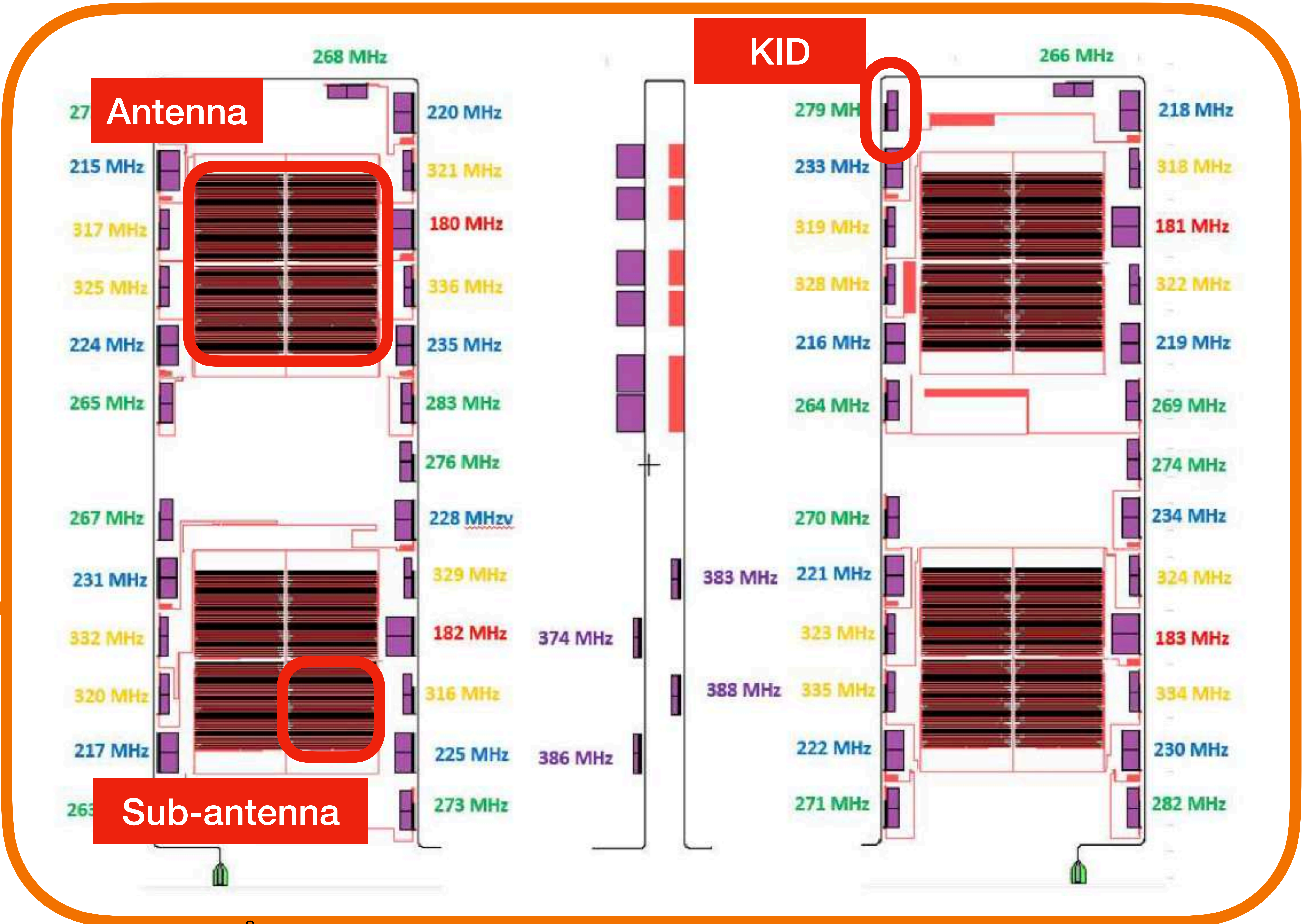
We can measure single photon event !

P. K. Day, H. G. LeDuc, B. A. Mazin, A. Vayonakis, and J. Zmuidzinas, “A broadband superconducting detector suitable for use in large arrays,” Nature, vol. 425, no. 6960, pp. 817–821, 2003.

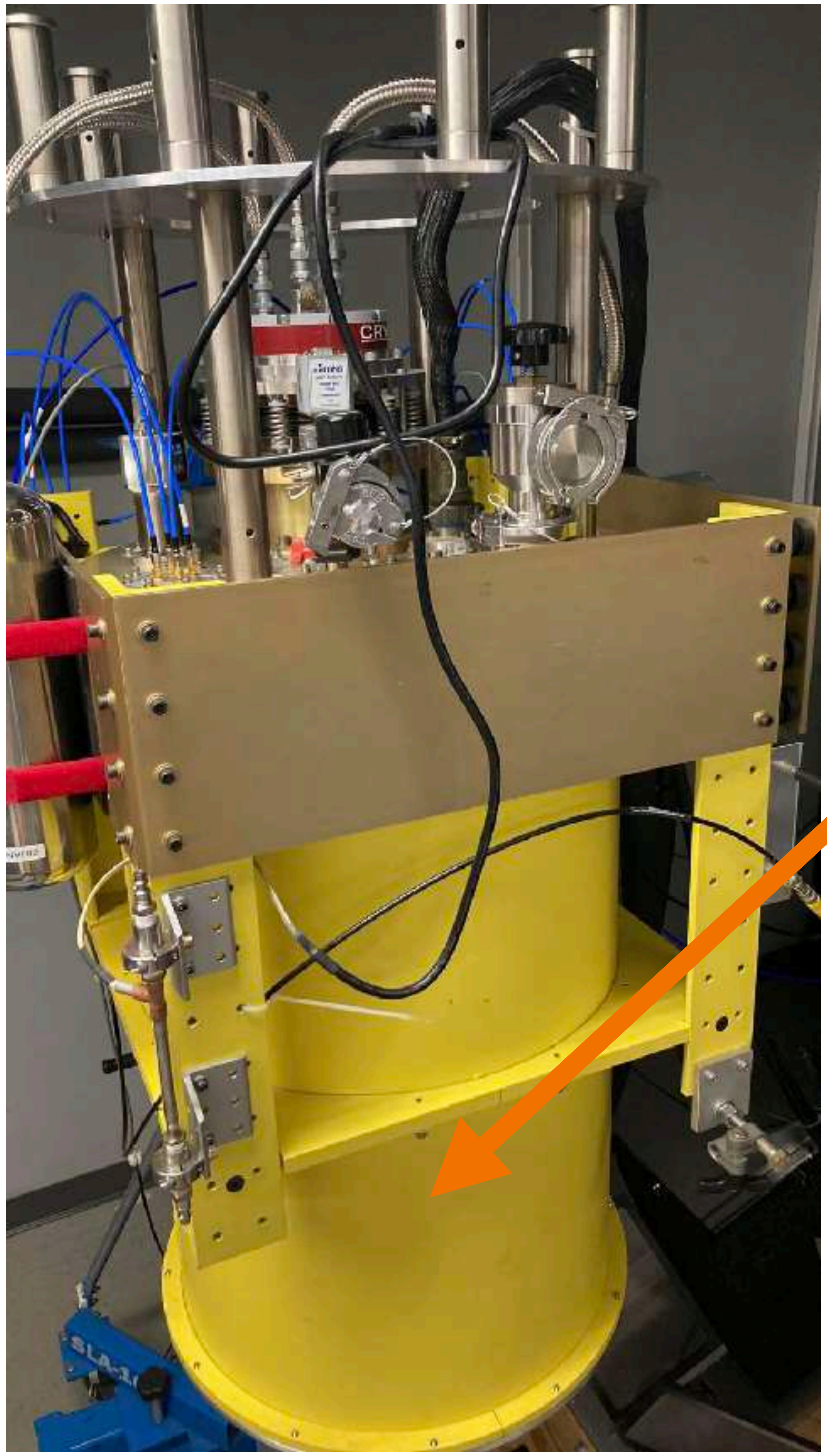
From MUSIC to NEW-MUSIC

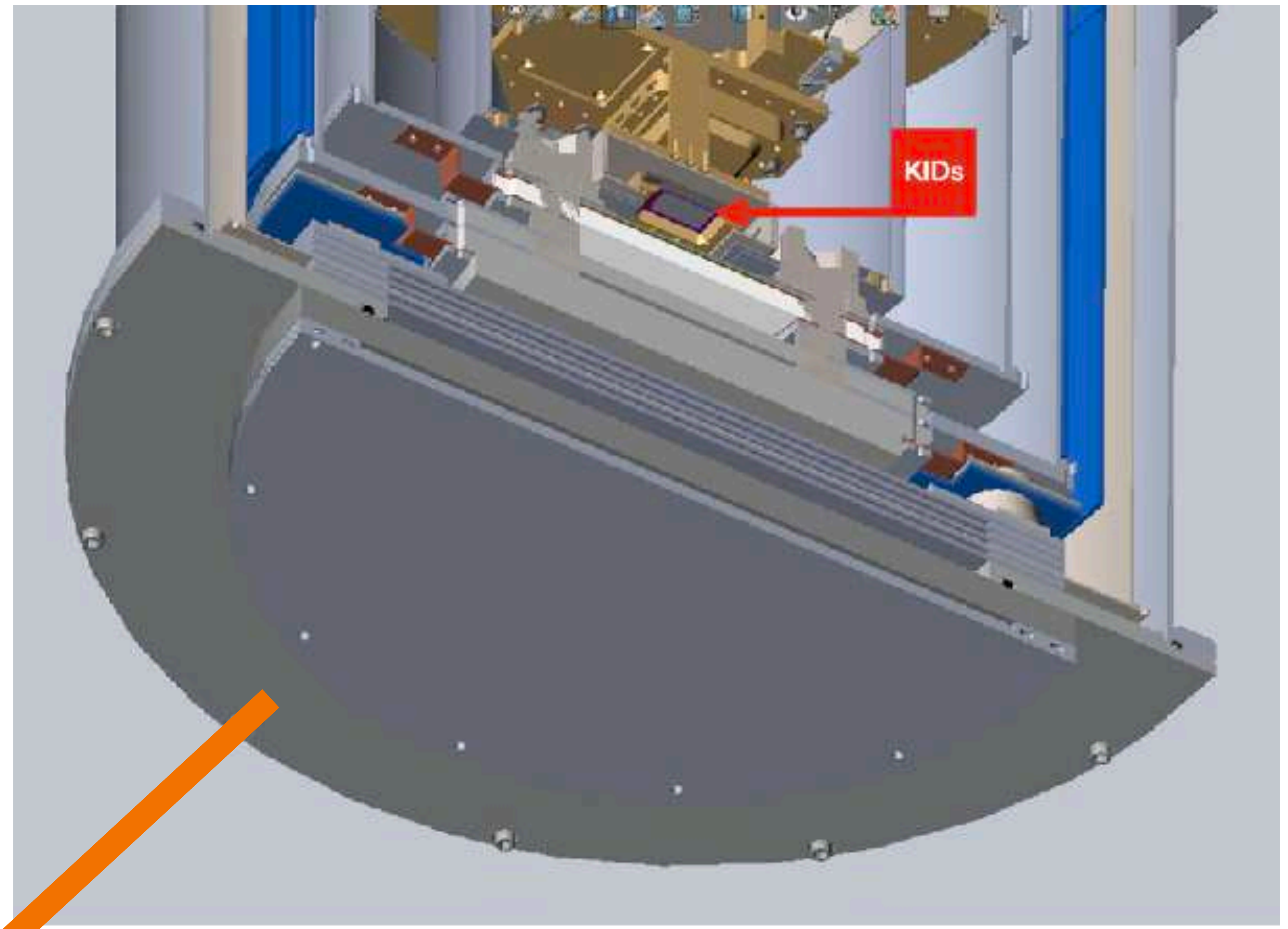
Caltech Submilimeter Observatory (Hawaii)

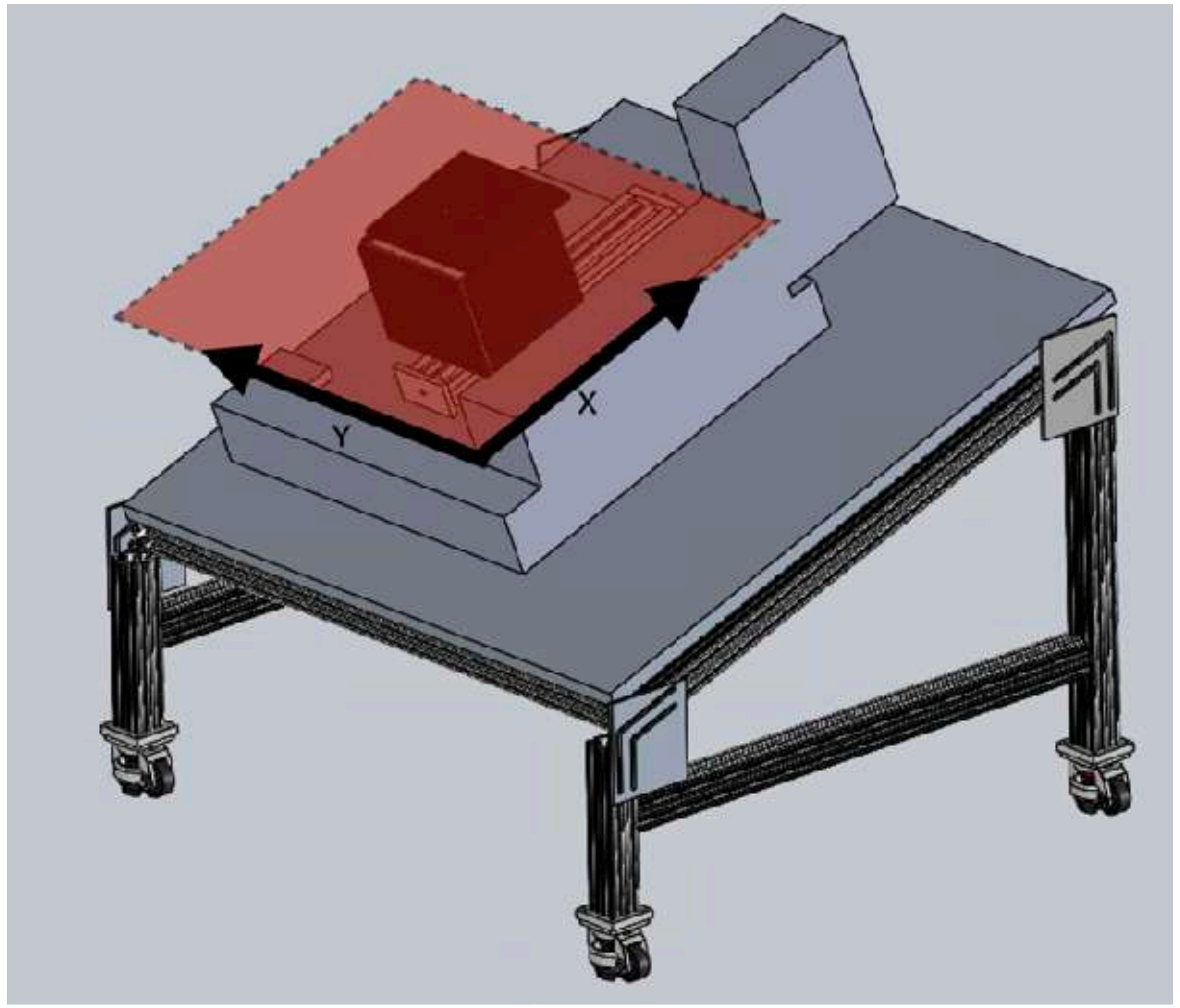




Objective of the project

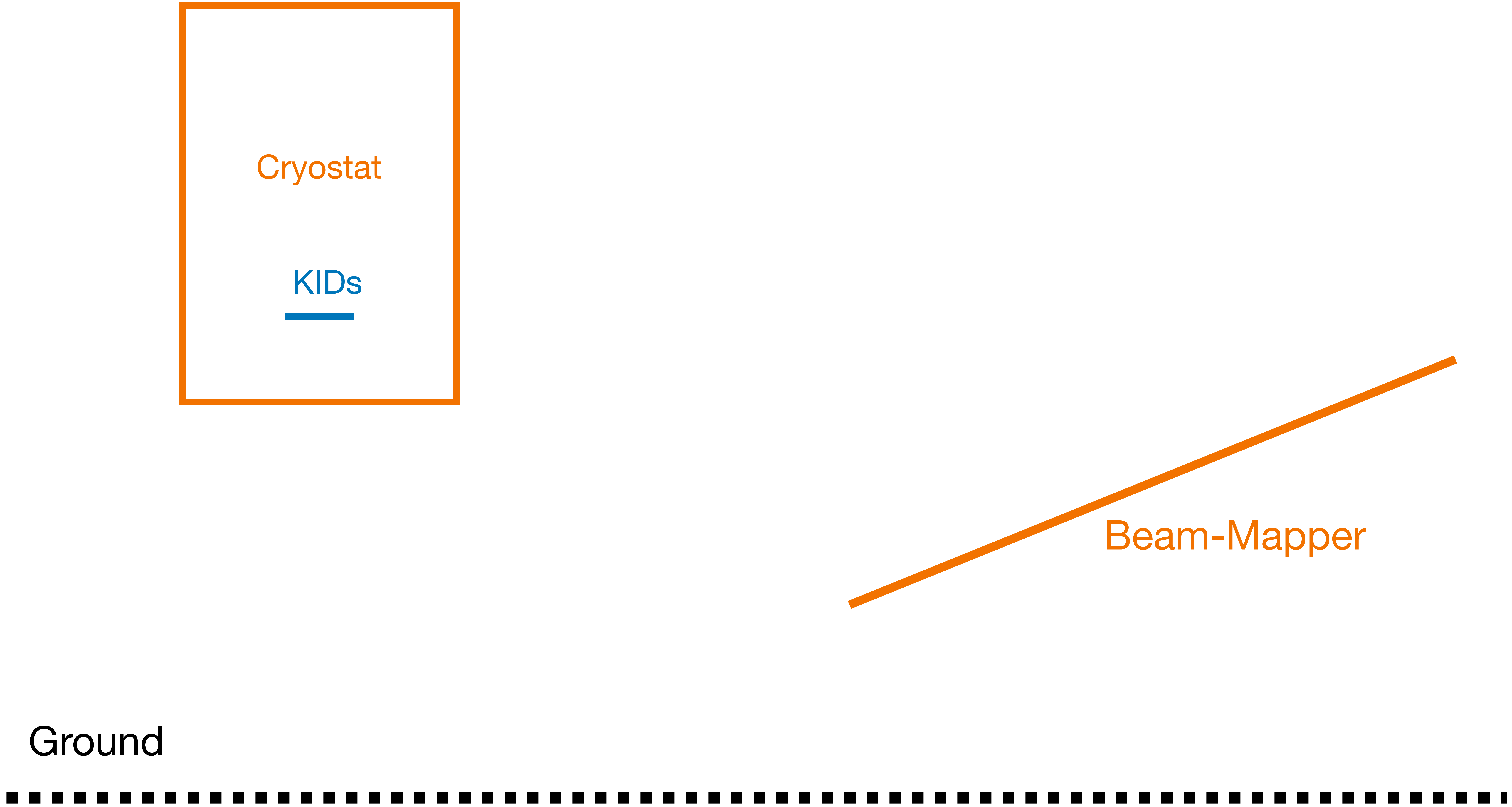




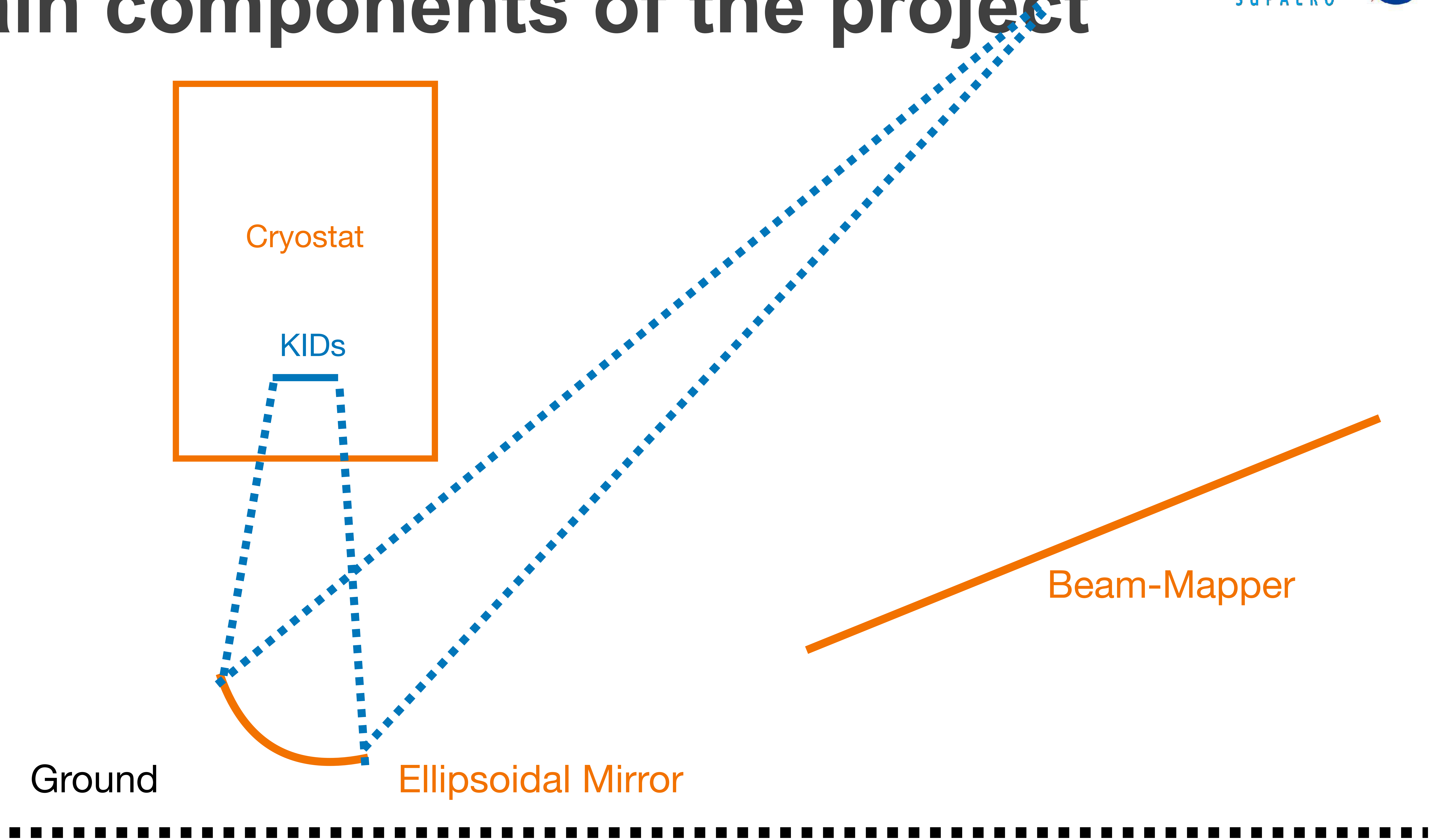


- Characterize the performance of new generations of detectors
- Designing an optical structure in order to rebuild the image of the detectors outside the cryostat
- Mapping and Identifying Antennas and Sub-antennas outside the cryostat

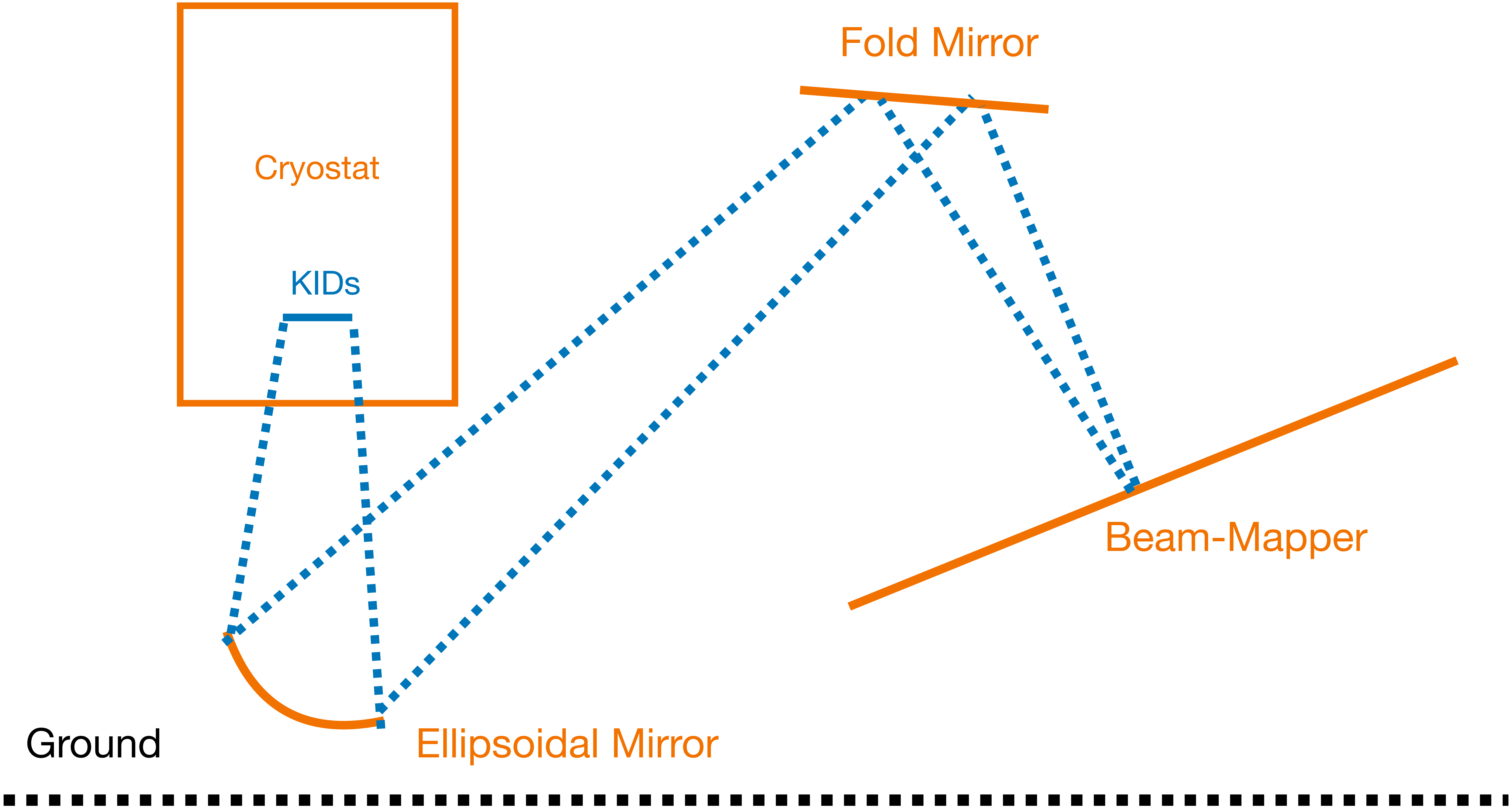
Main components of the project



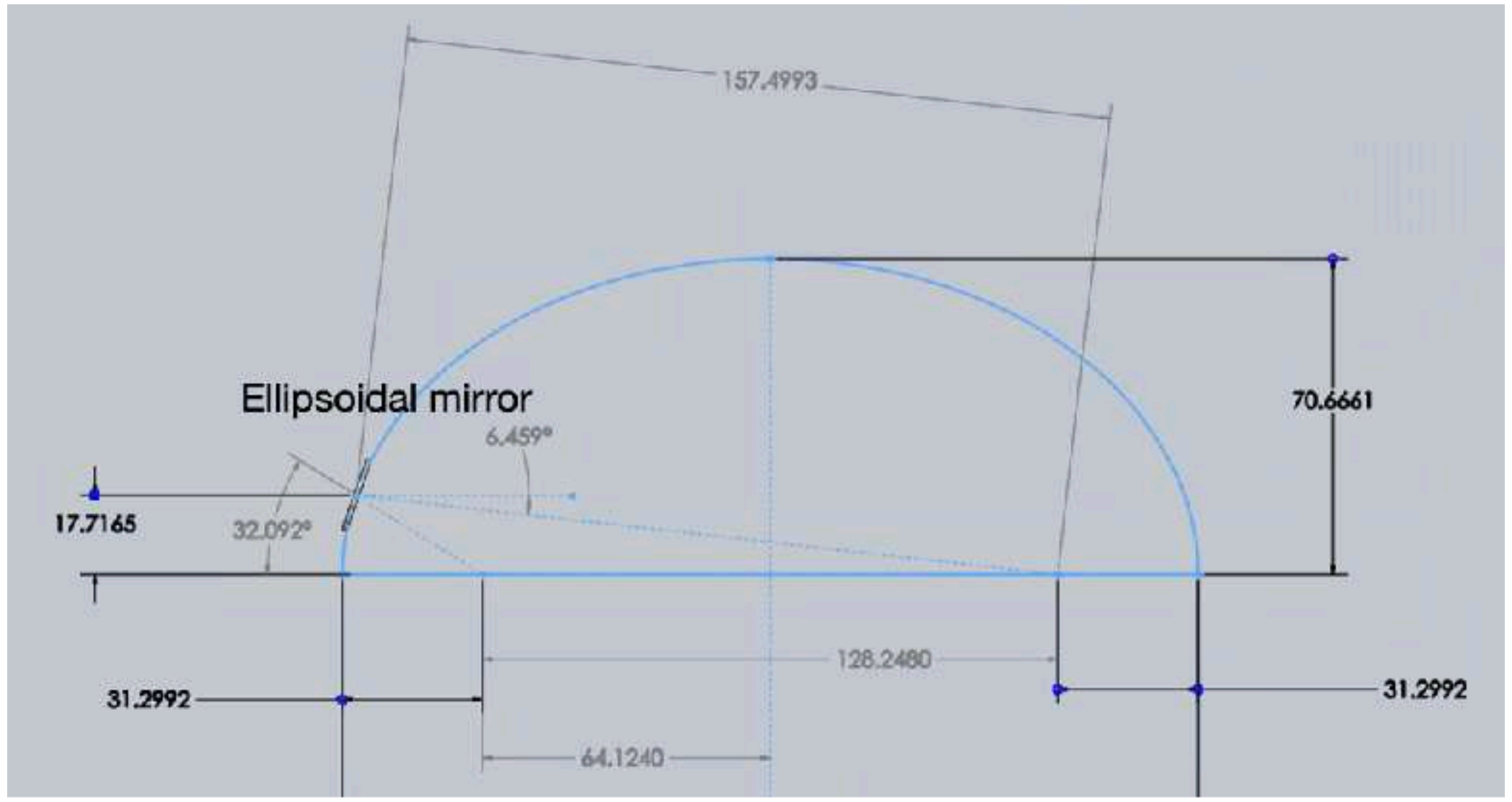
Main components of the project

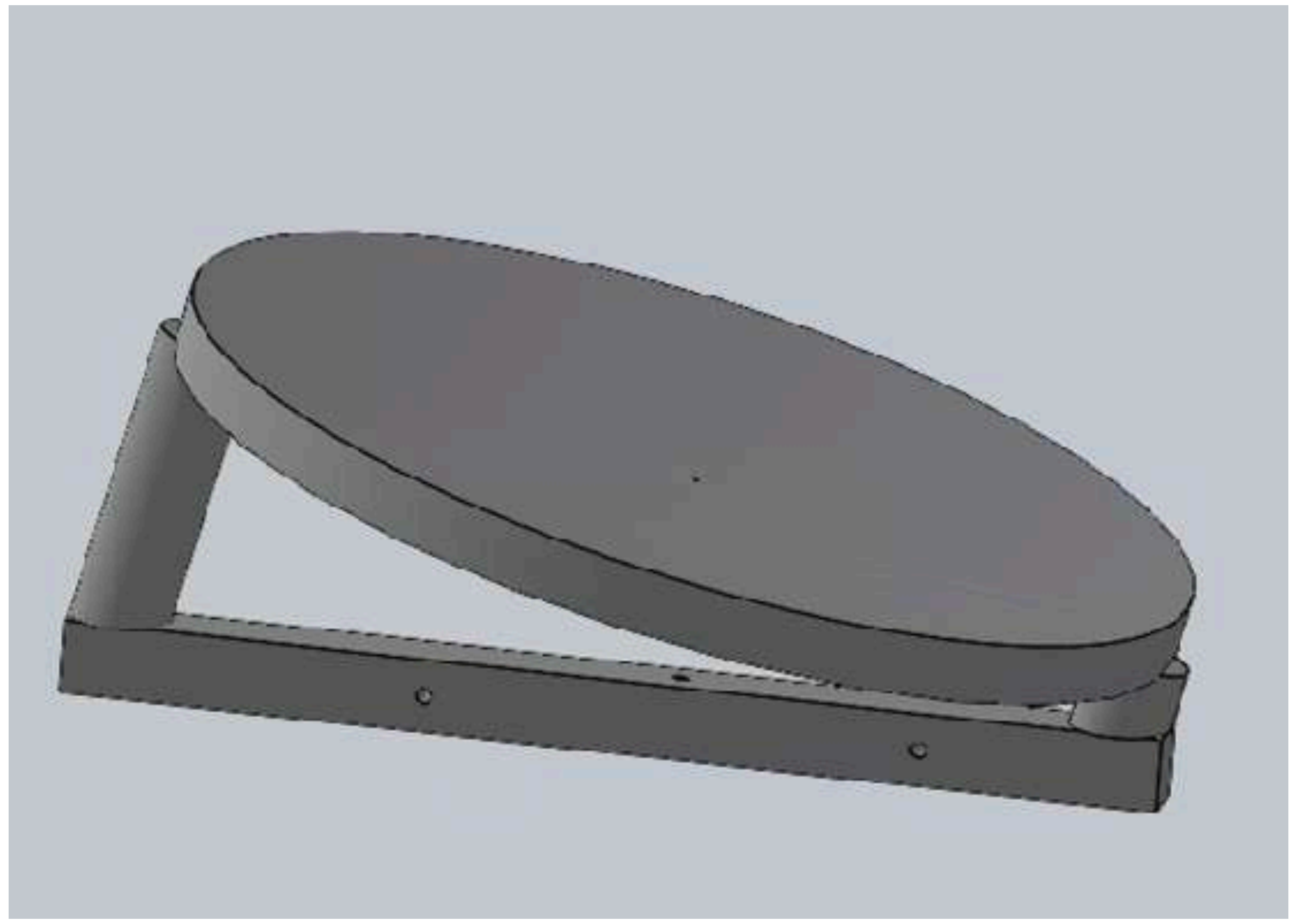


Main components of the project



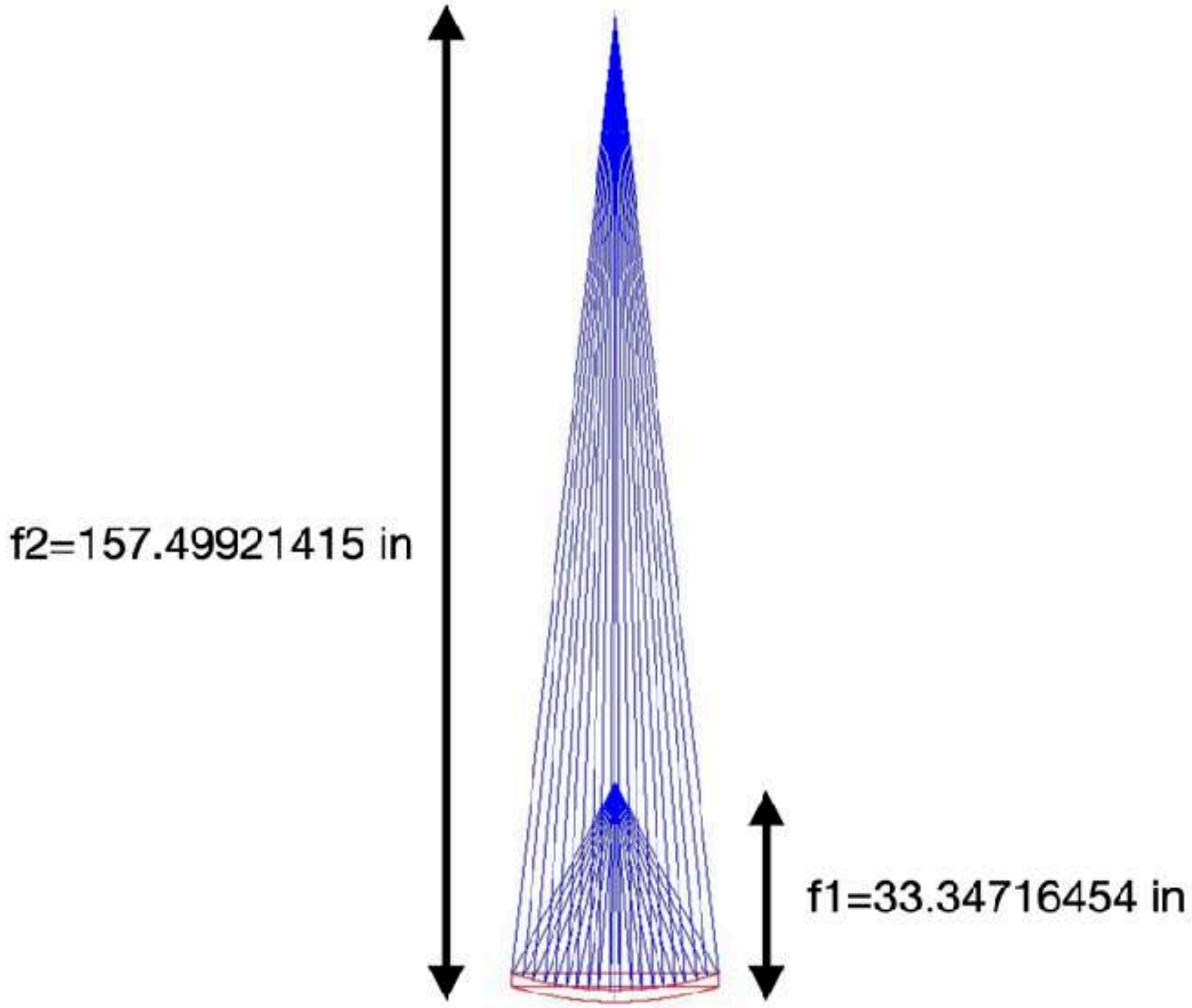
Ellipsoidal Mirror





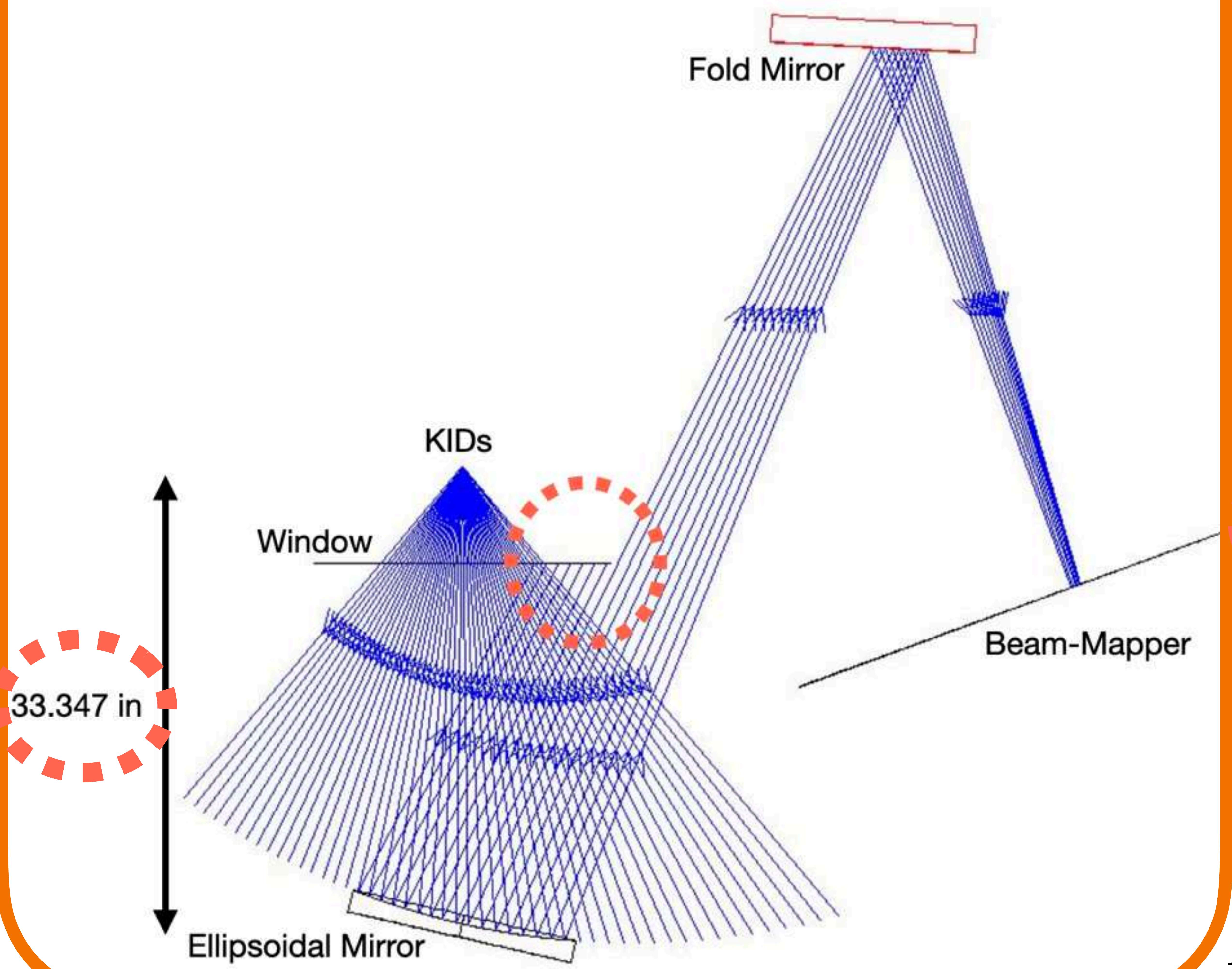
CAD	
Semi-major axis:	$a = 95.42in$
Semi-minor axis:	$b = 70.66in$
Eccentricity:	$e = \frac{\sqrt{a^2 - b^2}}{a}$

ZEMAX	
Radius of curvature:	$R = 54.643in$
Conic Constant:	$K = -0.823$

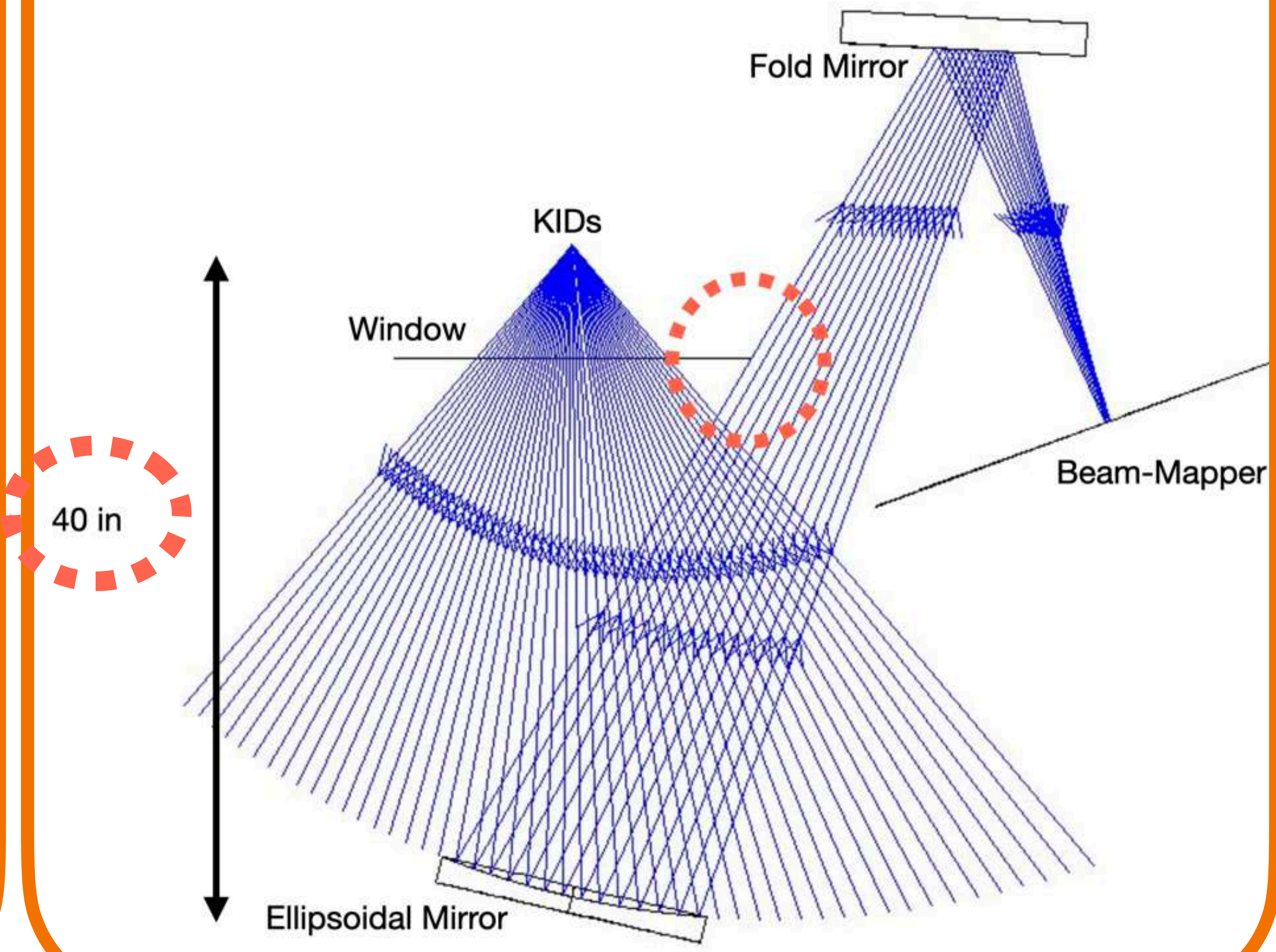


Zemax Simulation

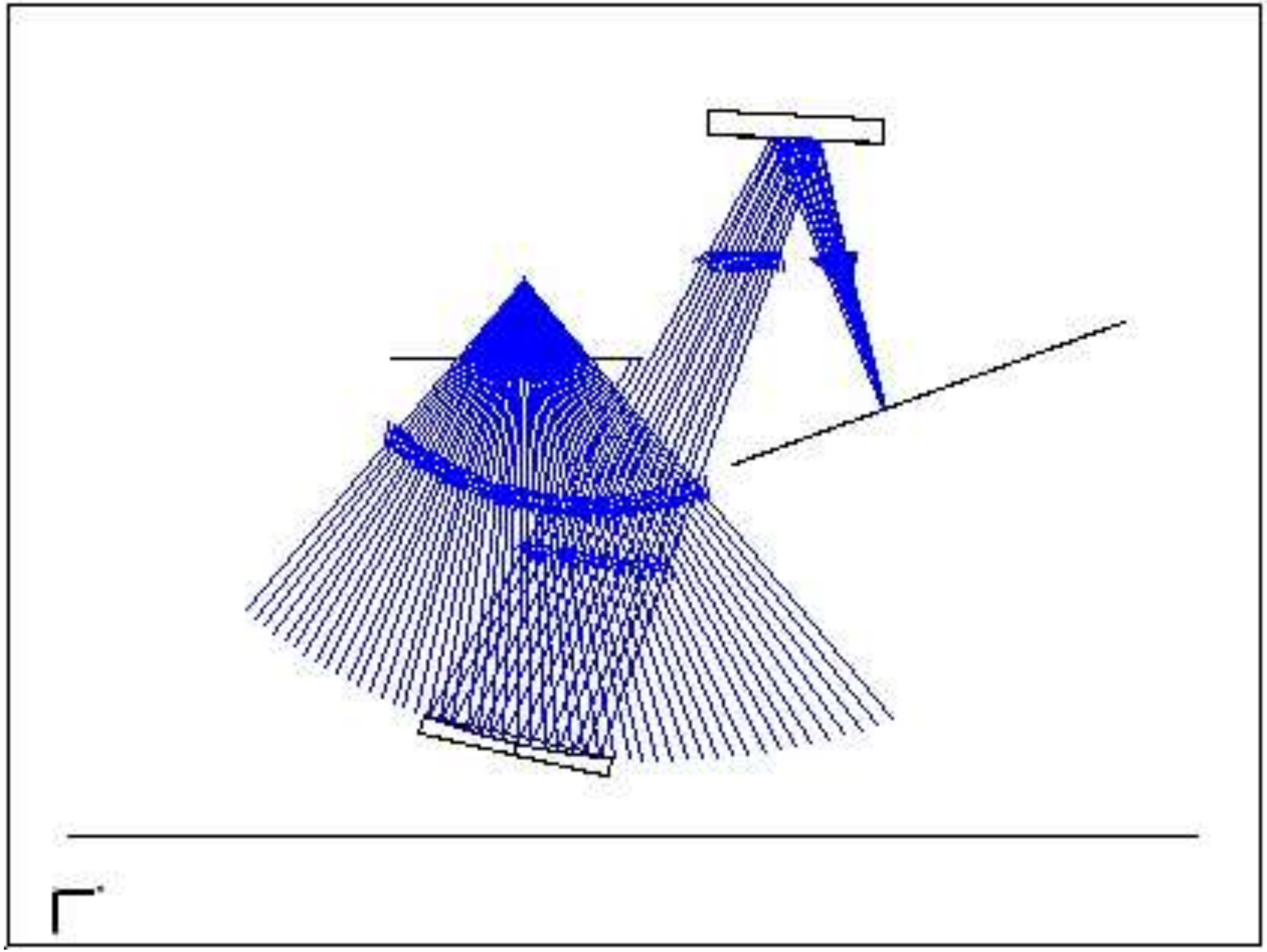
Distance KIDs - Mirror = f1



Distance KIDs - Mirror > f1

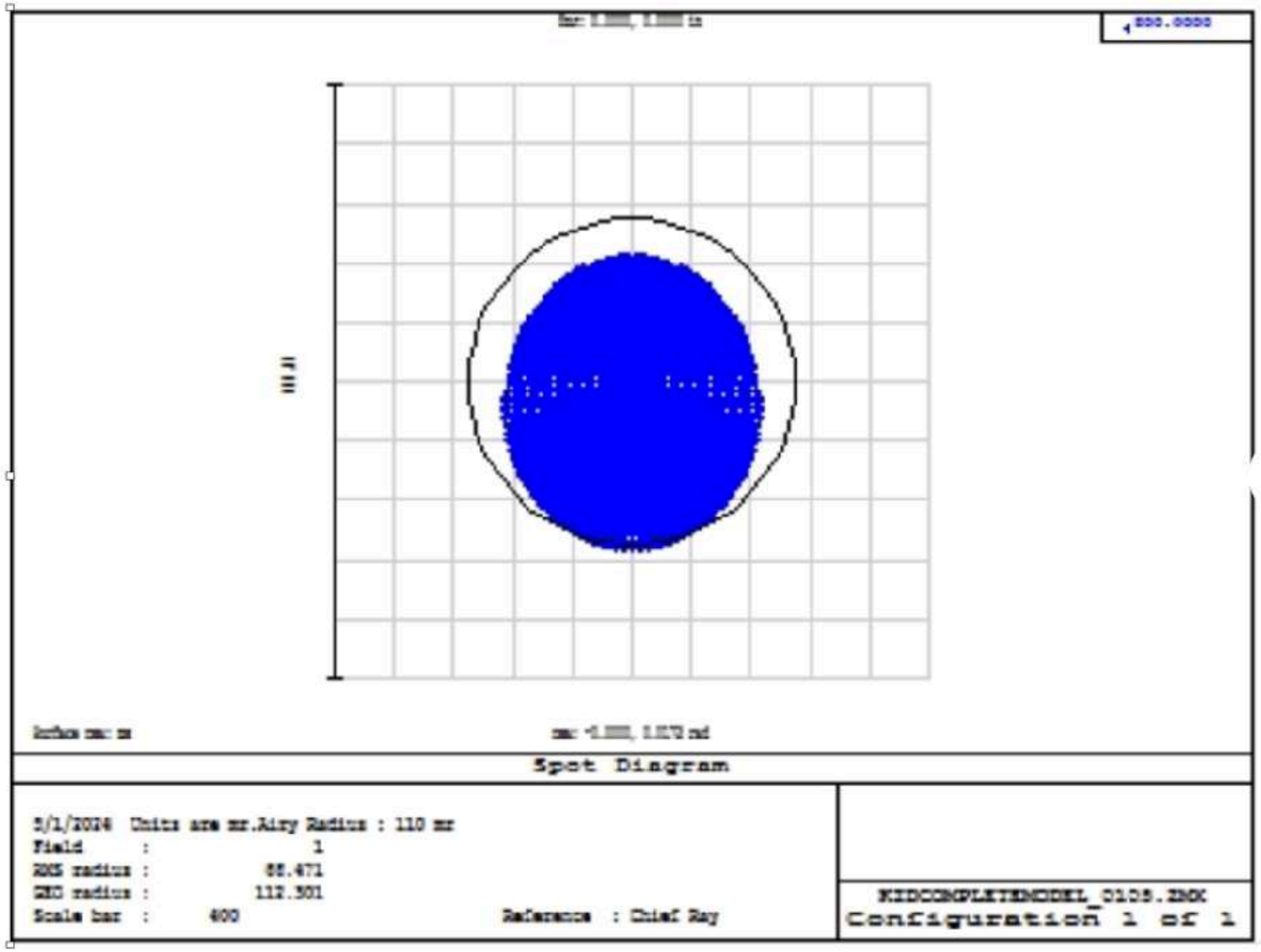


Zemax Simulation

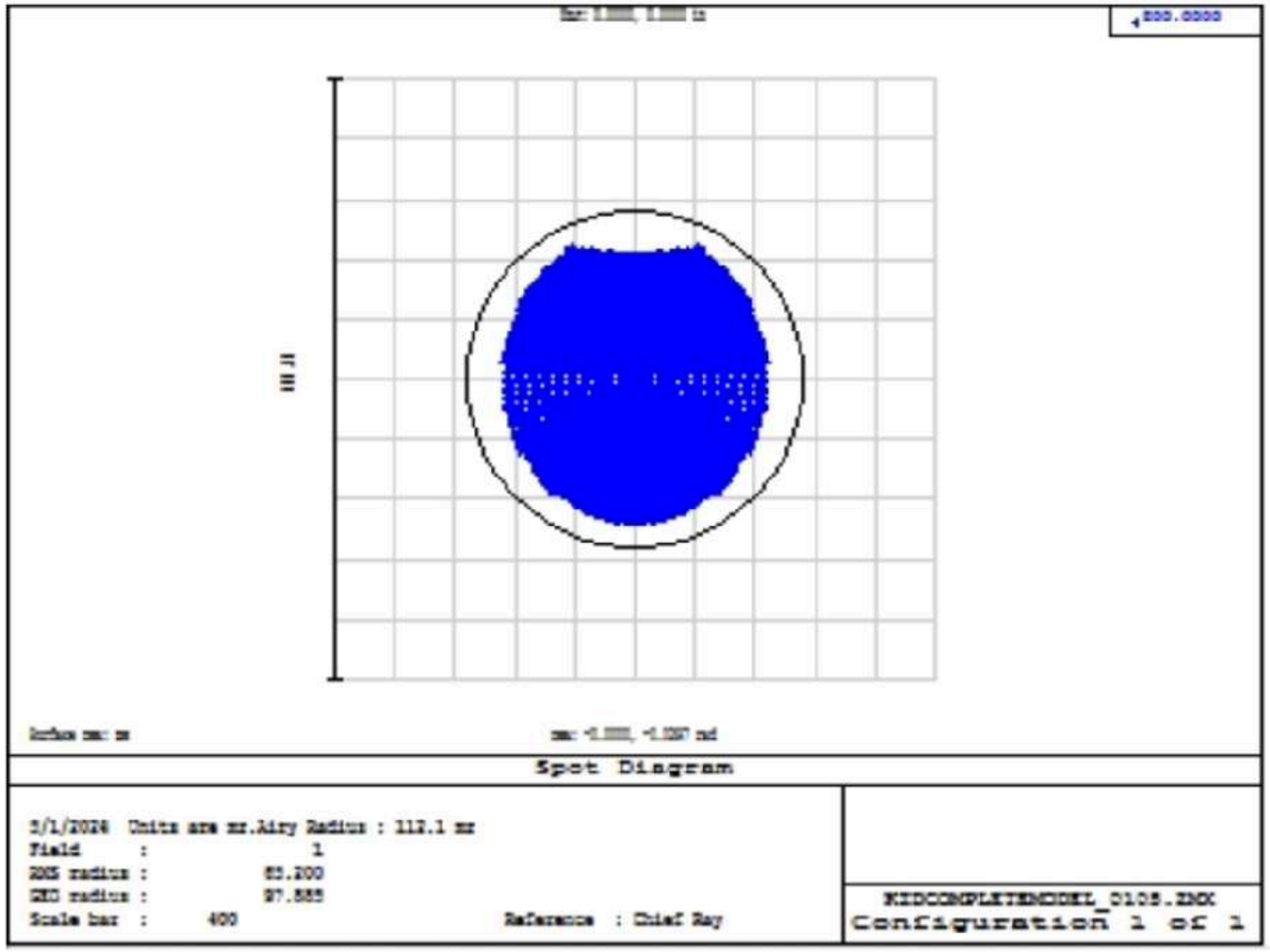


Zemax Simulation - Airy Disks

KIDs at -3.3mm

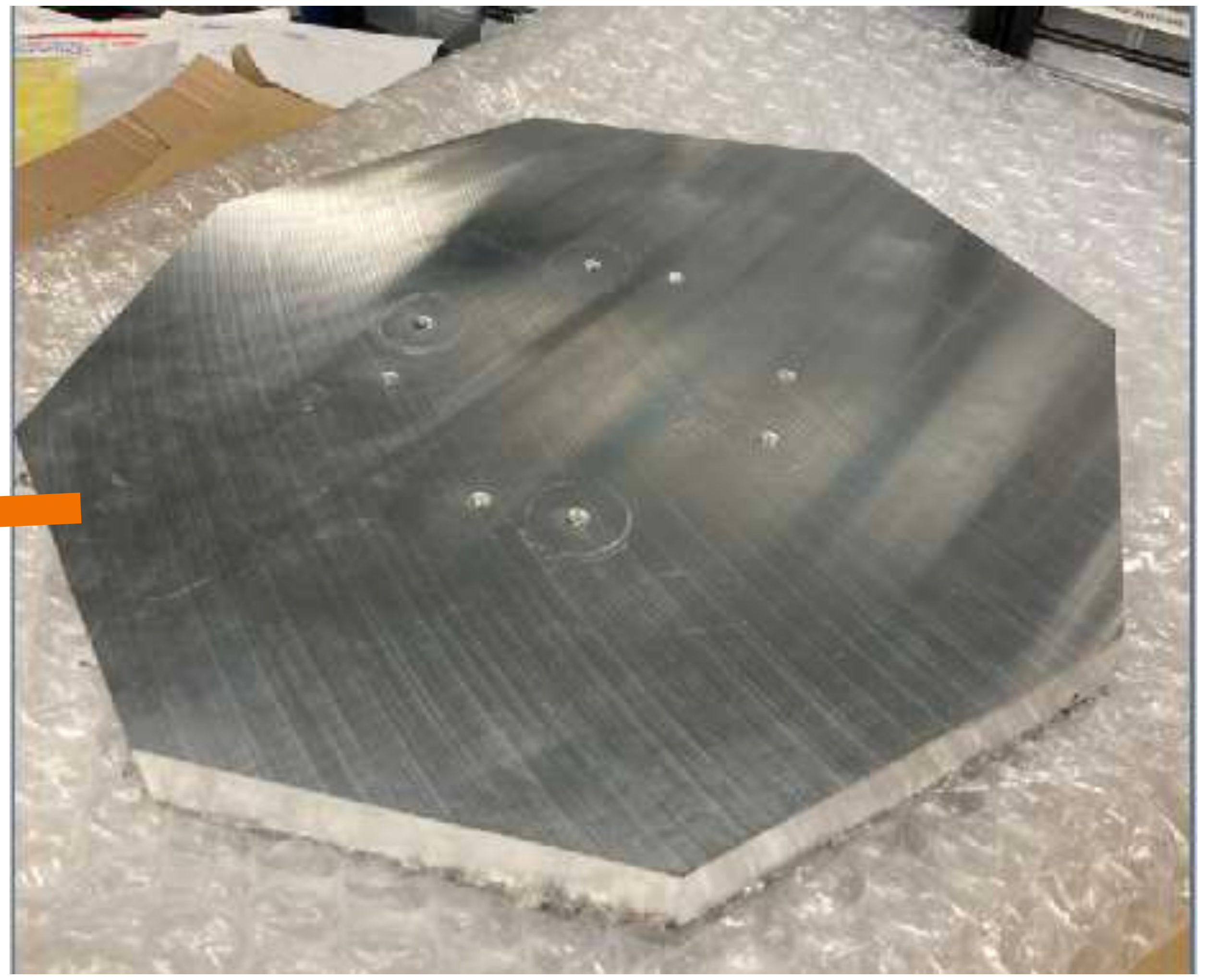


KIDs at +3.3mm

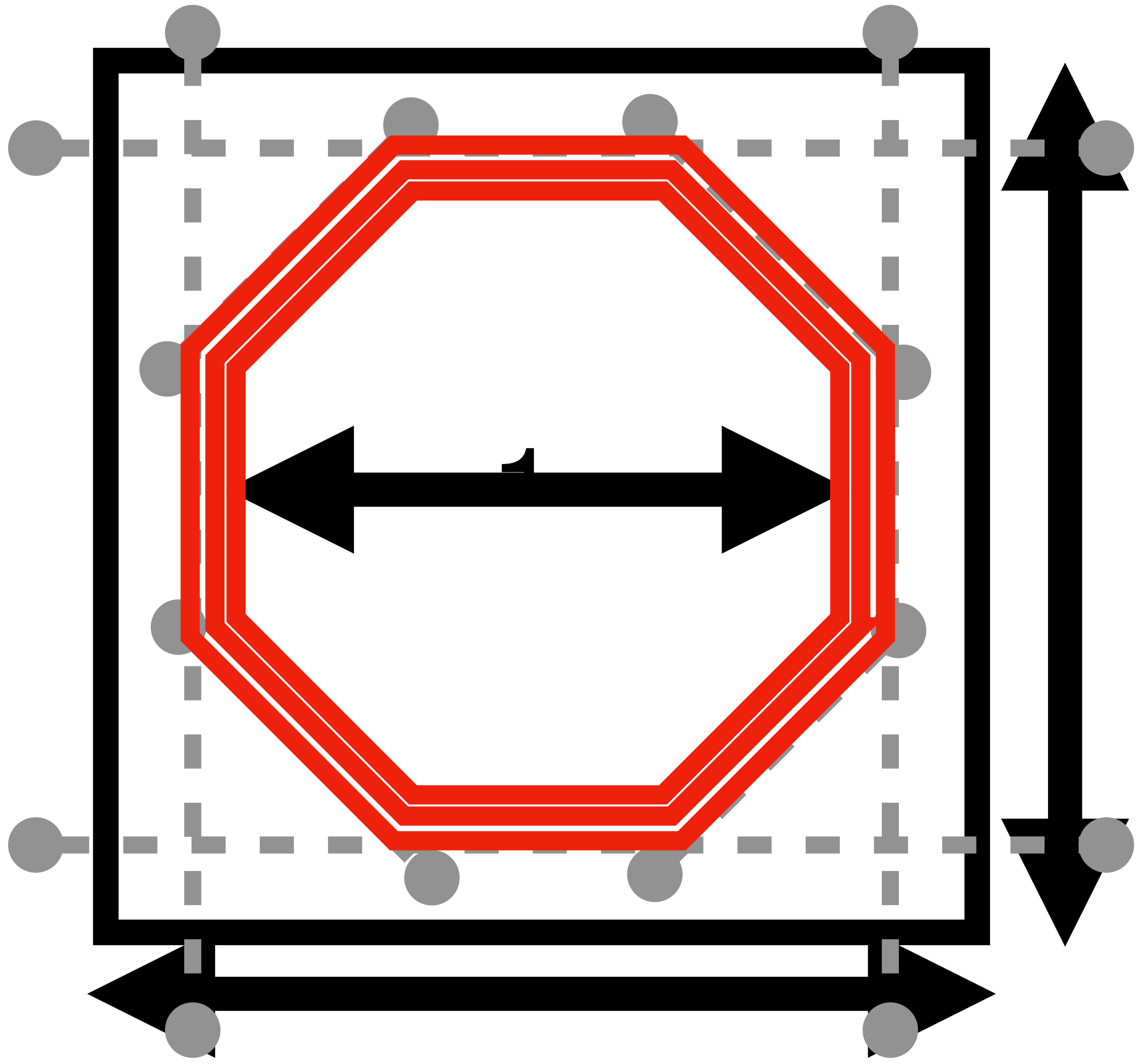


Design CAD

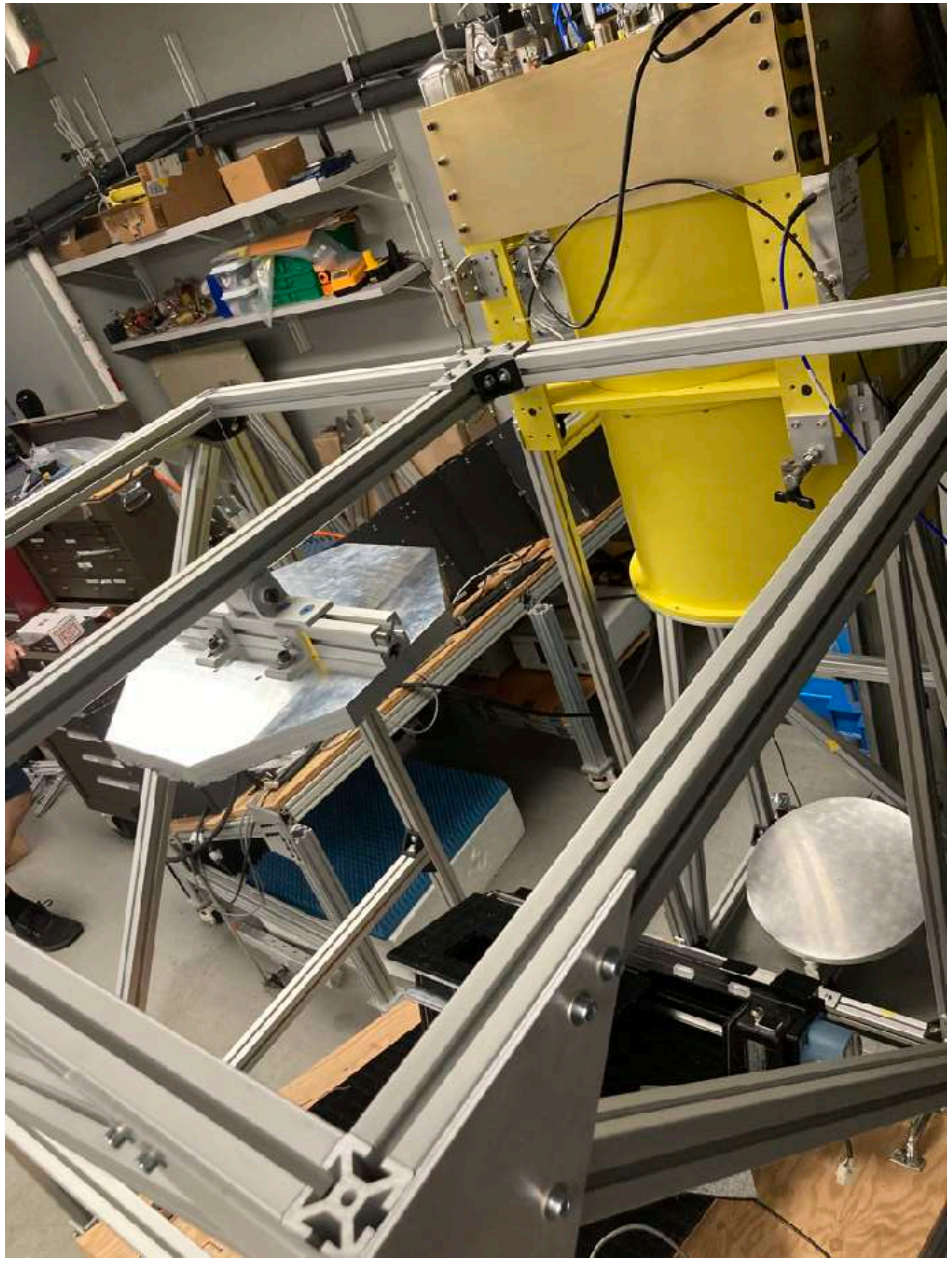
Construction



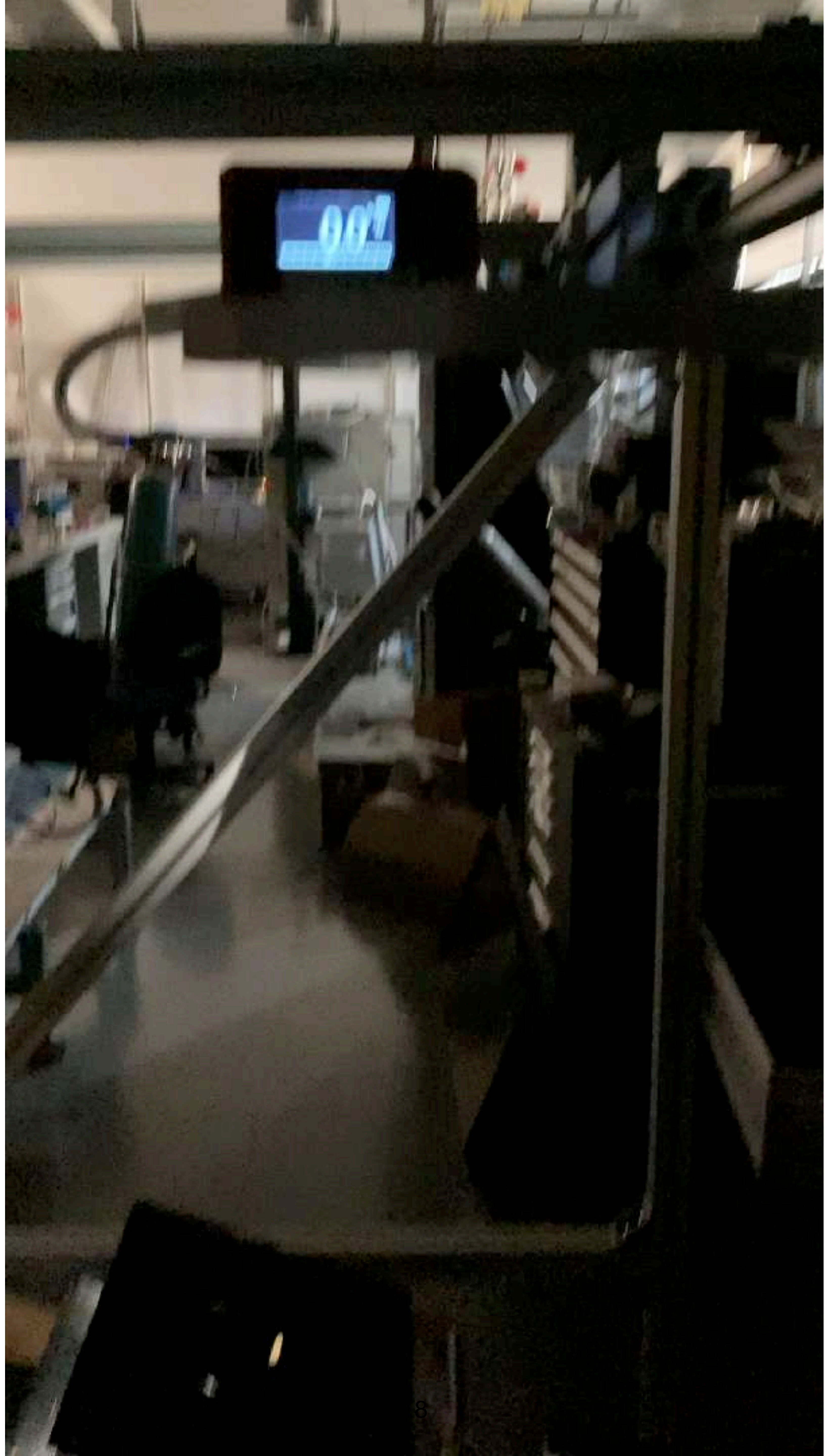
Fold Mirror



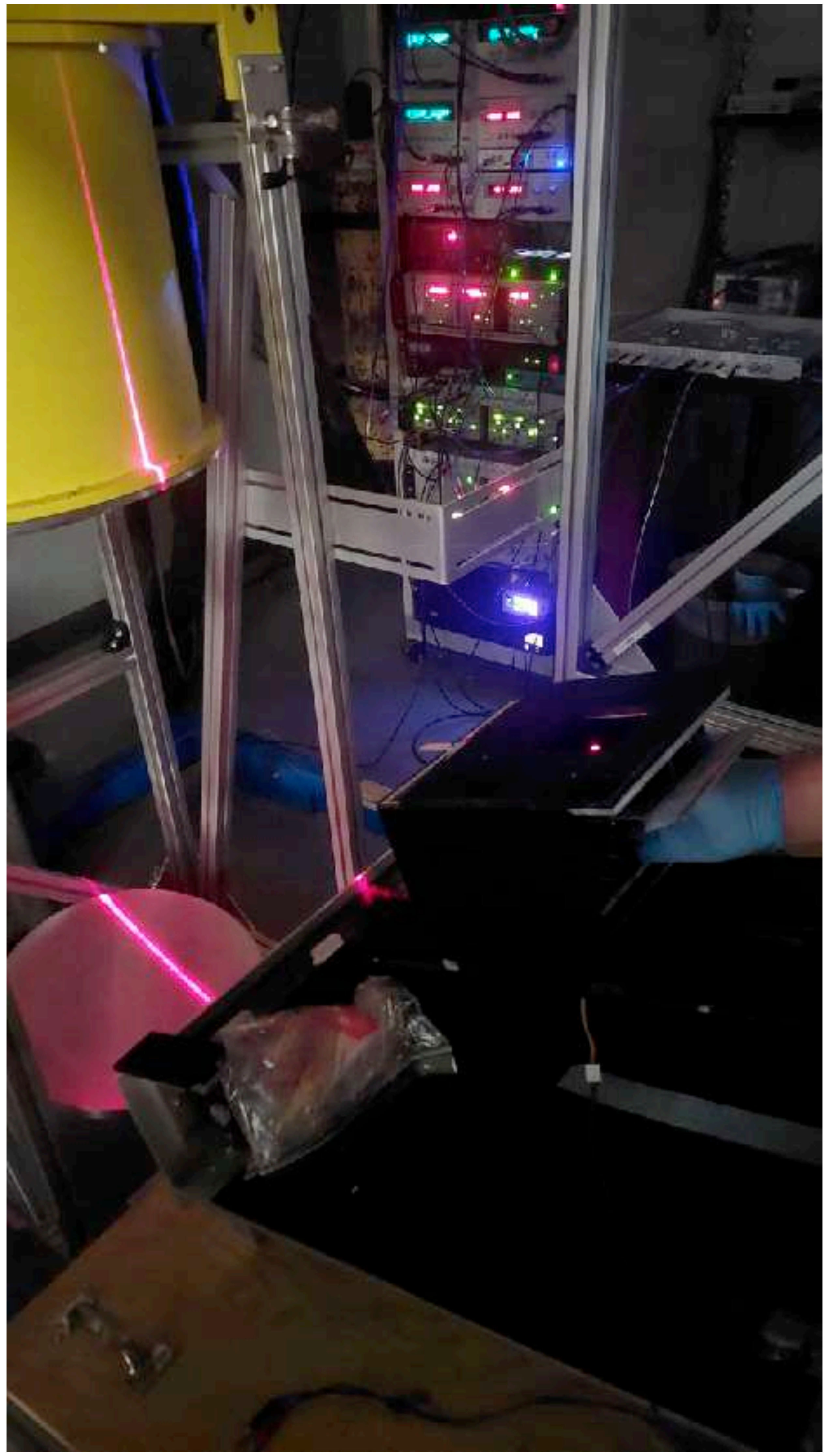
Construction

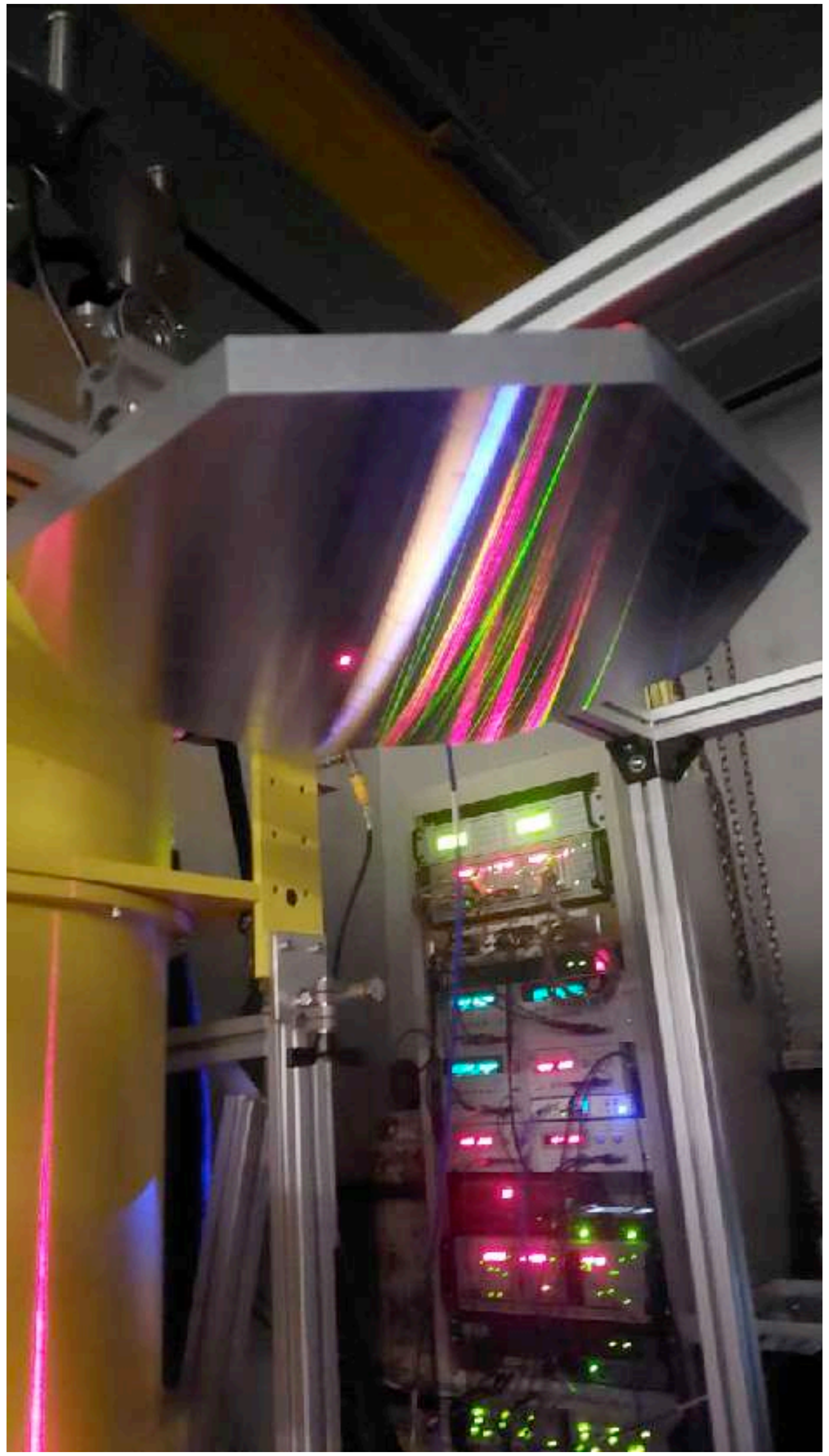


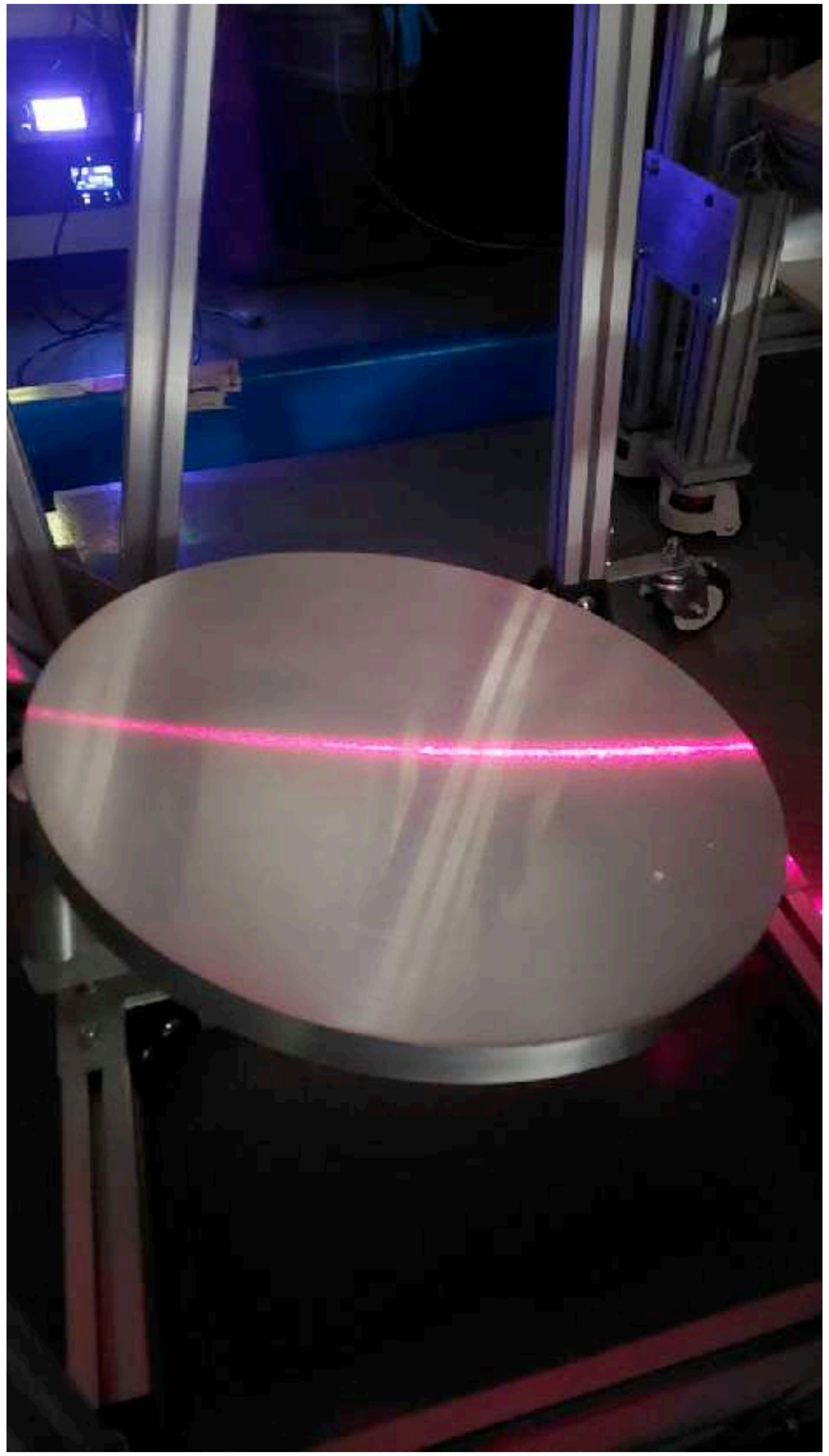
Measures

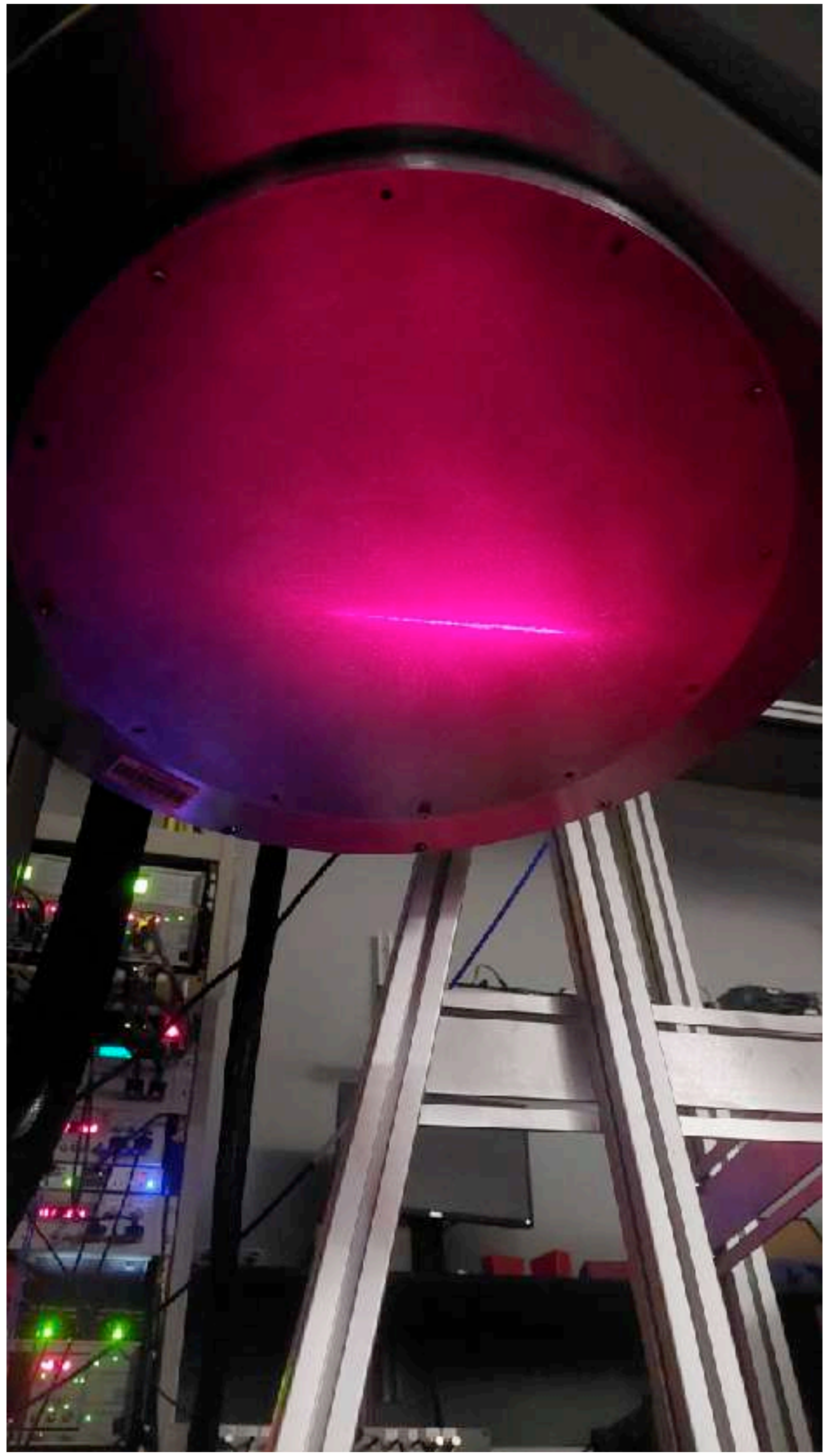


Measures

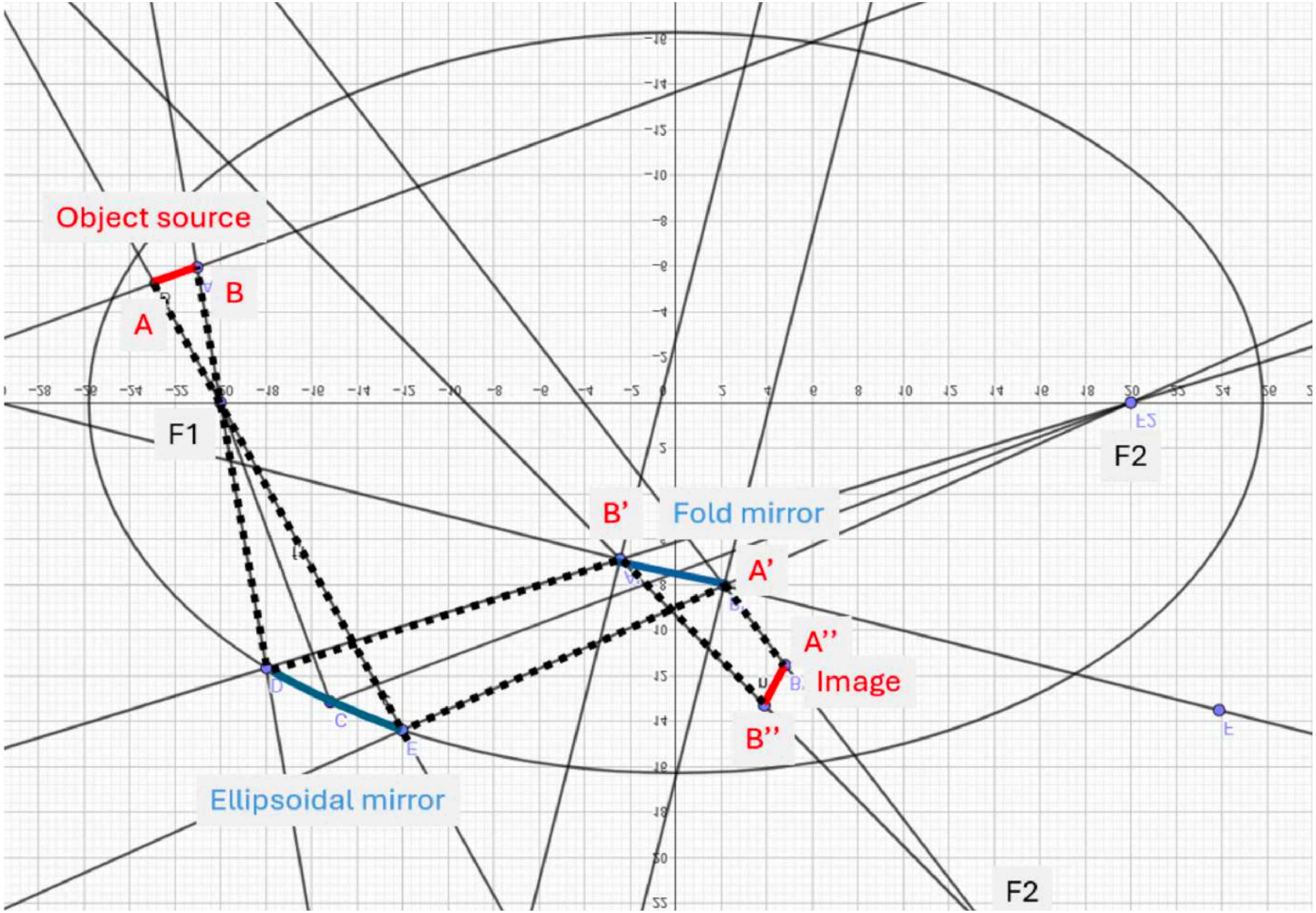




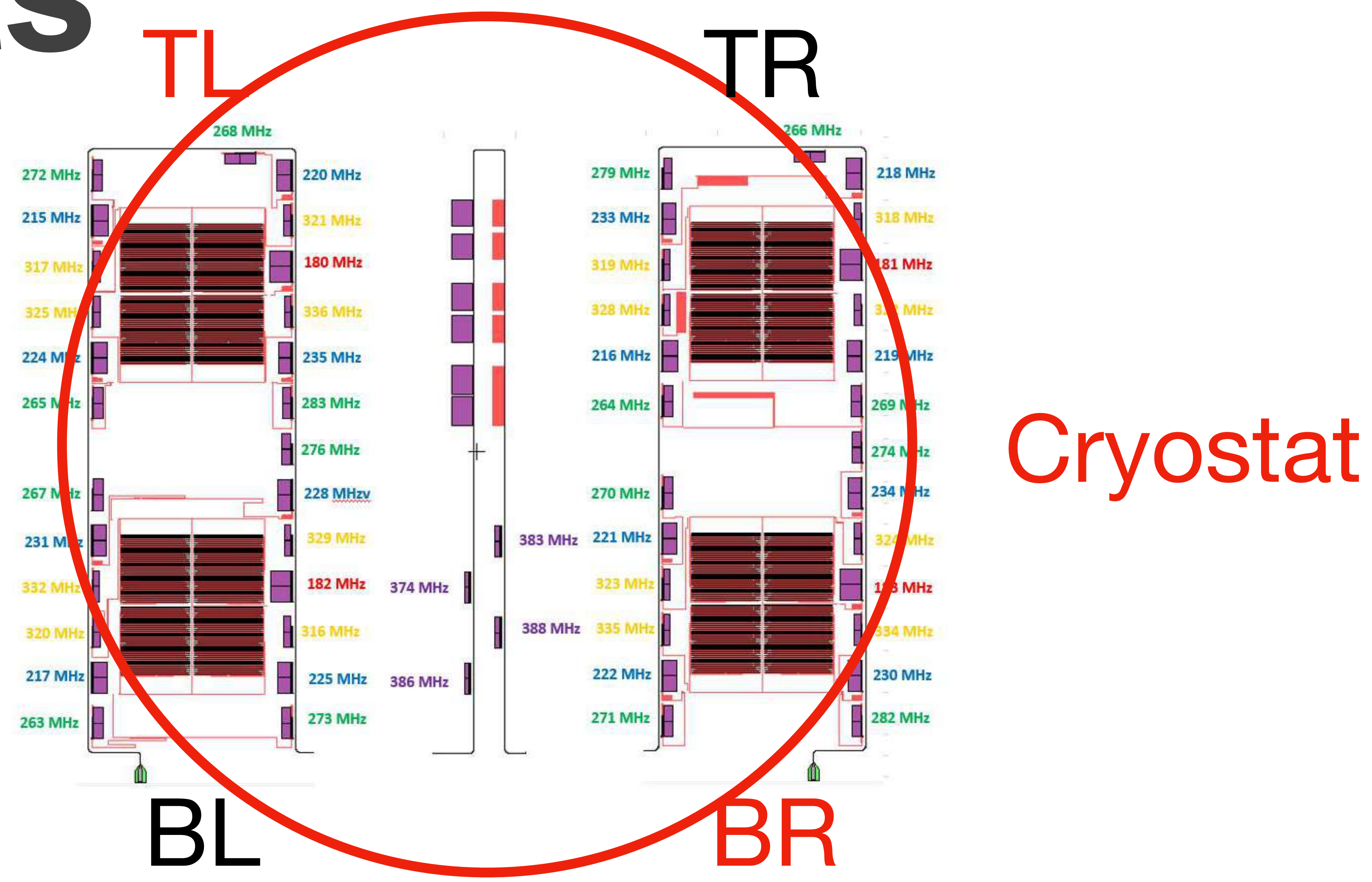




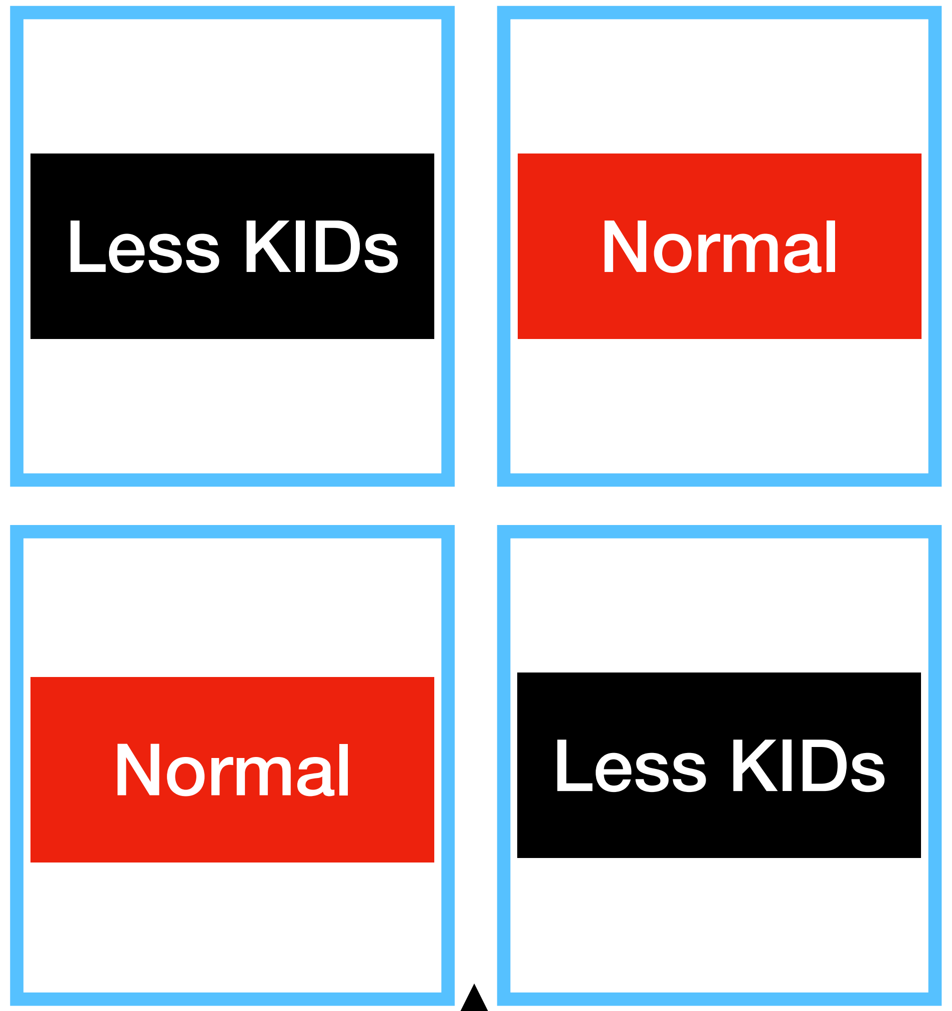
Results


Object source
A
B
F1
F2
B'
Fold mirror
A'
A''
Image
B''
Ellipsoidal mirror
F2

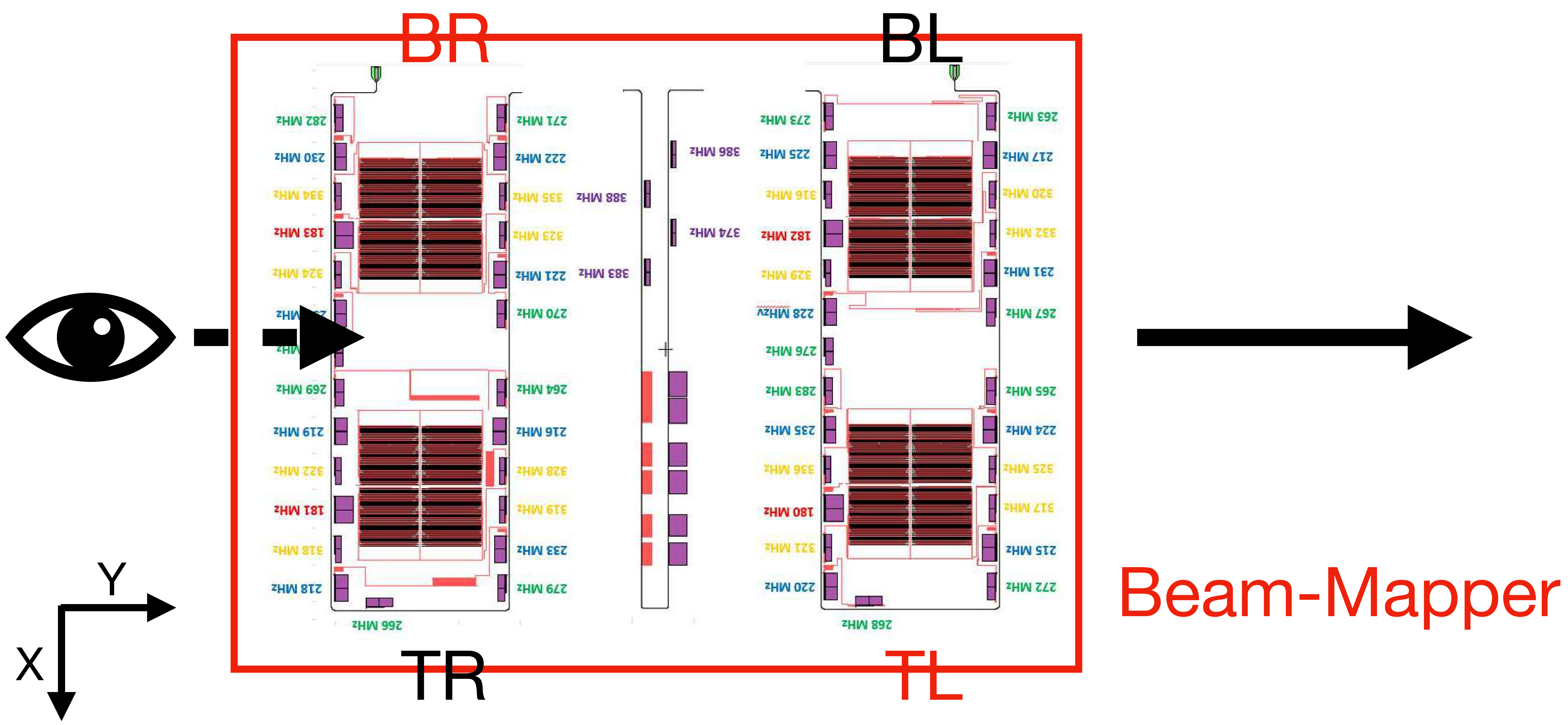
Results



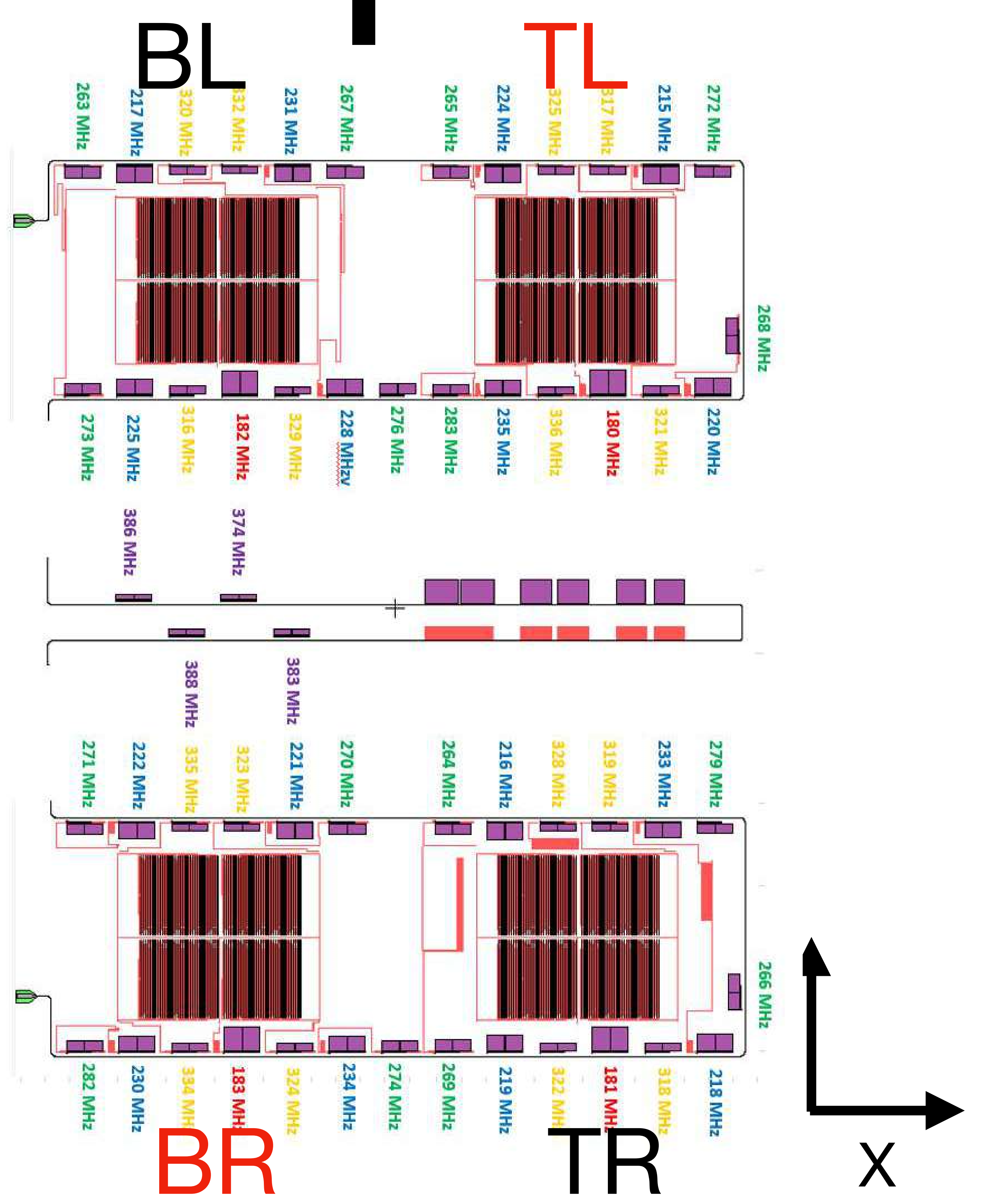
TL
TR
Cryostat
BL
BR



Less KIDs
Normal
Normal
Less KIDs

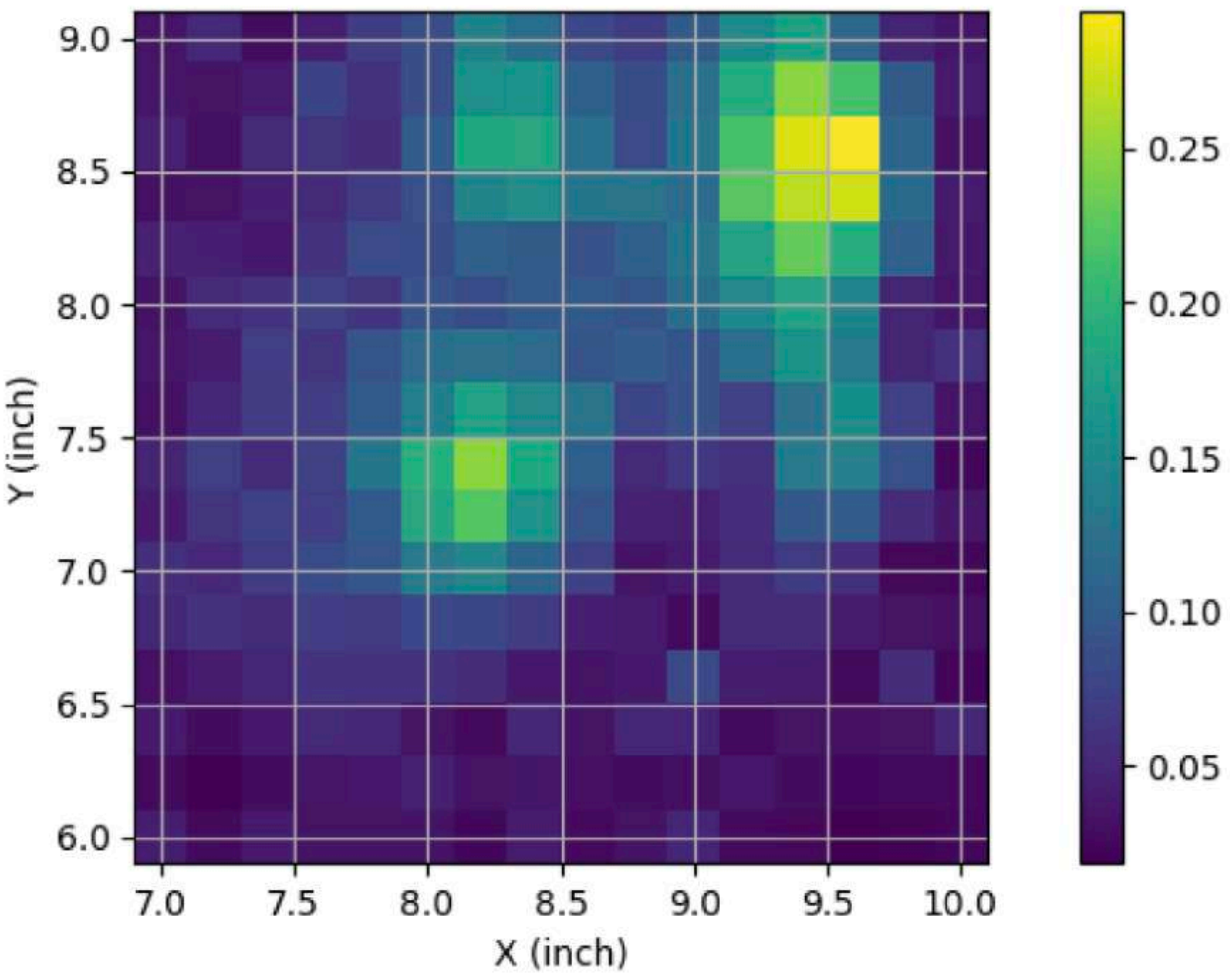


BR
BL
Beam-Mapper
TR
TL
Y
X

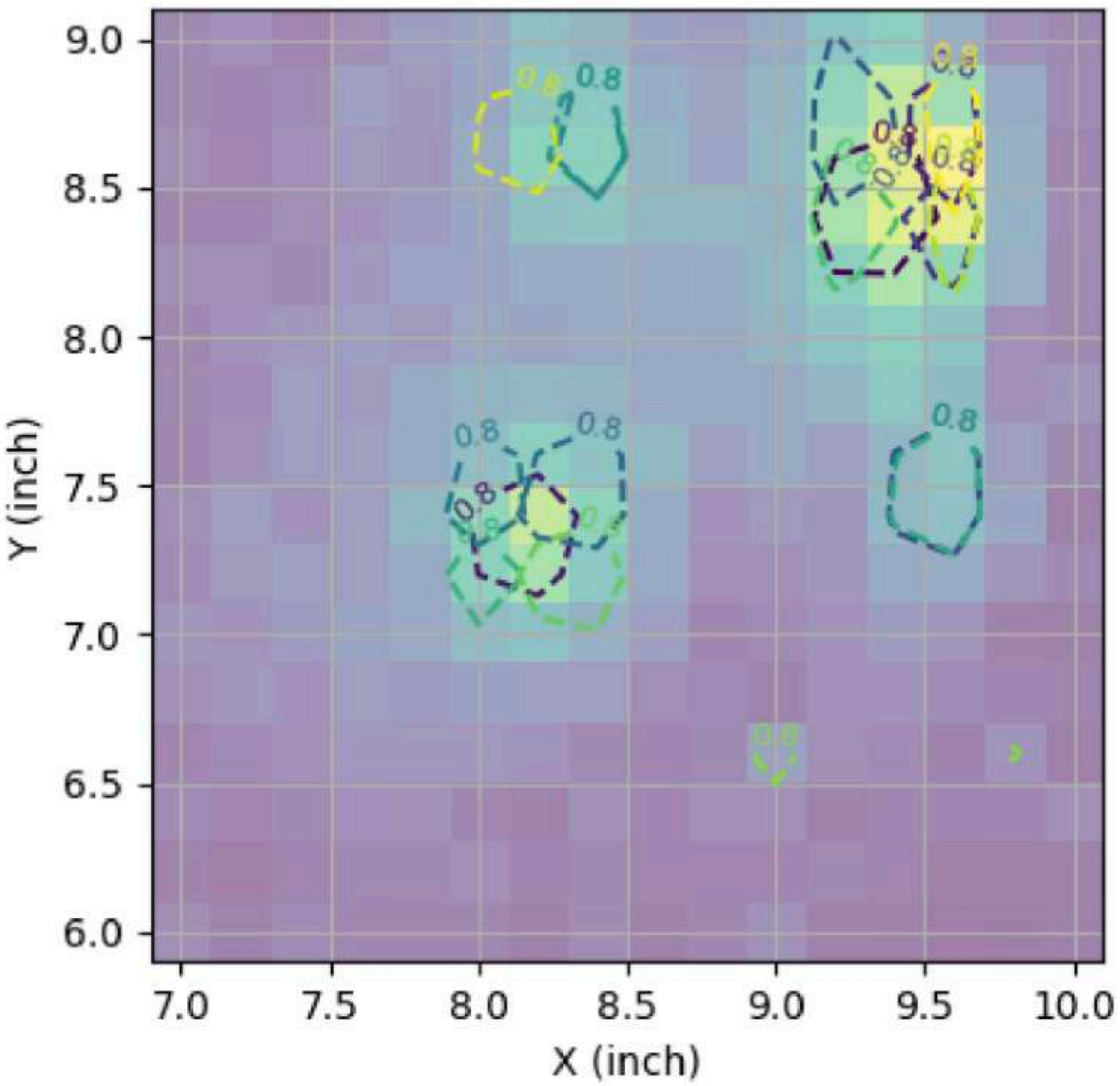


BL
TL
BR
TR
X

Measures

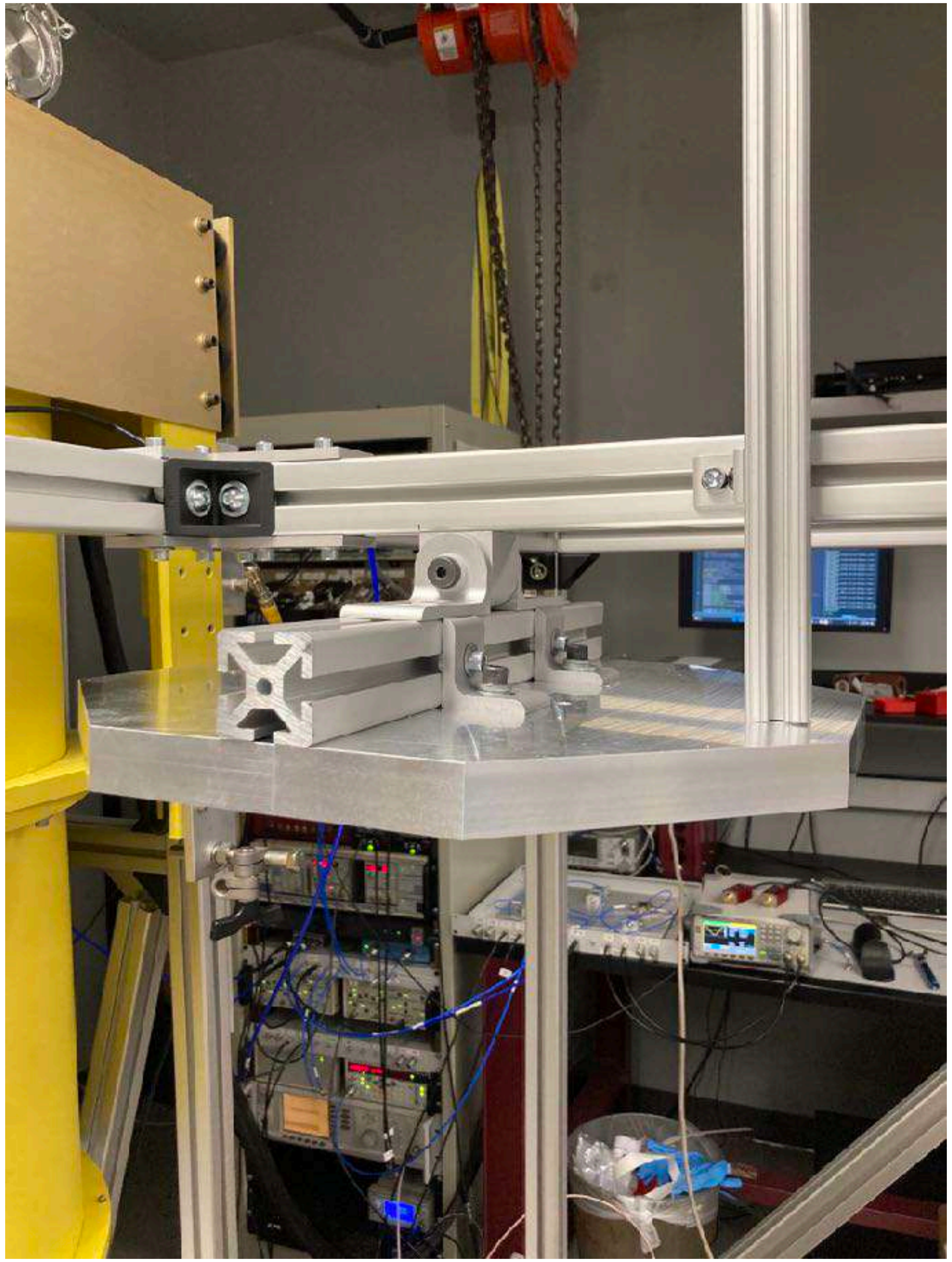


9.0
8.5
8.0
7.5
7.0
6.5
6.0
Y (inch)
7.0 7.5 8.0 8.5 9.0 9.5 10.0
X (inch)
0.25
0.20
0.15
0.10
0.05

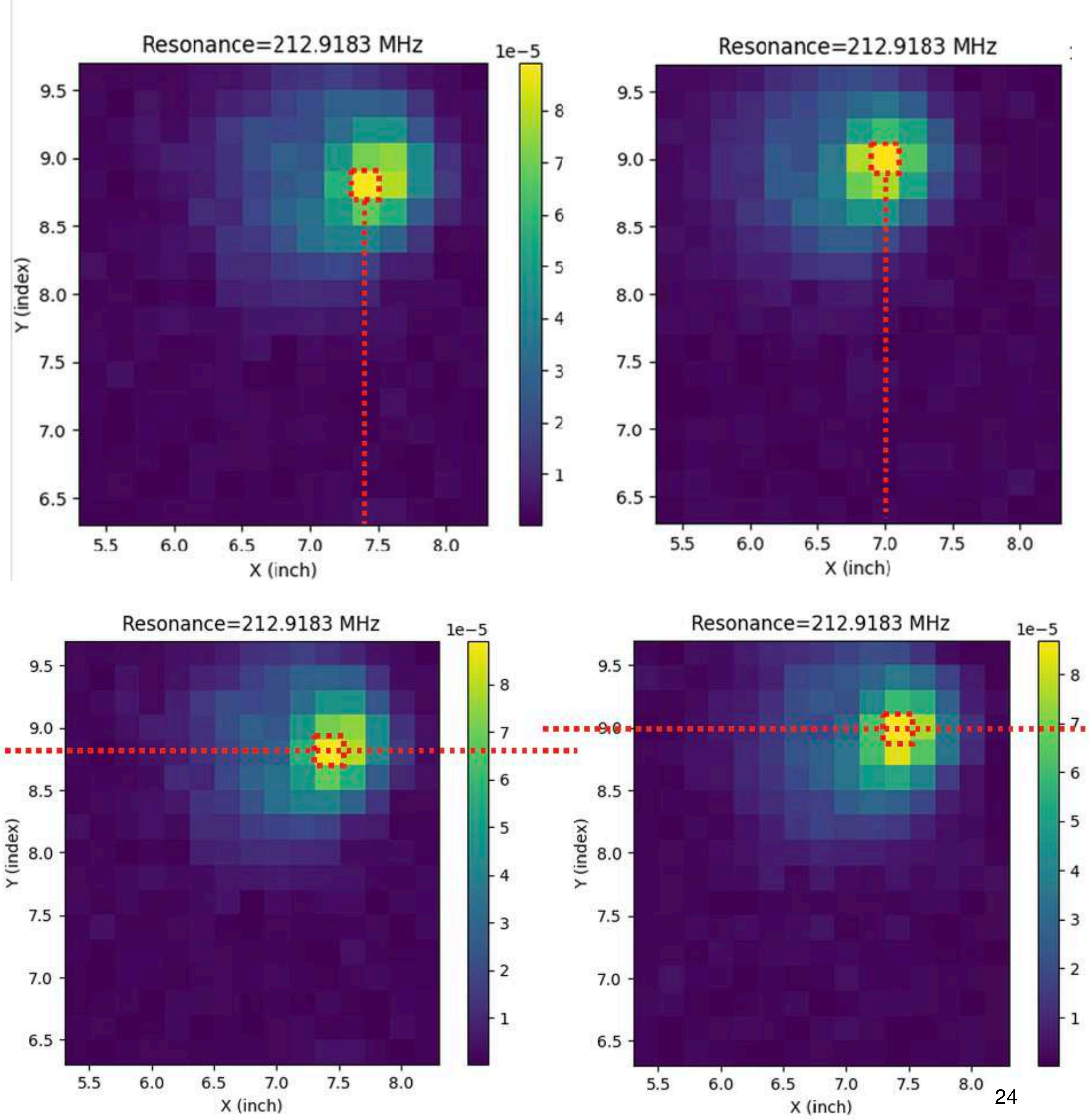


9.0
8.5
8.0
7.5
7.0
6.5
6.0
Y (inch)
7.0 7.5 8.0 8.5 9.0 9.5 10.0
X (inch)
0.8

Optimization structure



Reproducibility



Deviation
$\Delta X \approx 0.4in$
$\Delta Y \approx 0.2in$

Conclusion